

26 October 2022
Babies A & B
+ starting on Baby C

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Wednesday, 26 October 2022
(10.30 am)

MR JUSTICE GOSS: I thought you were going to say something?

MR JOHNSON: No. Sorry, I'm just a bit slow this morning.

MR JUSTICE GOSS: That's all right, Mr Johnson.

All right, jury in.

(In the presence of the jury)

DR SANDIE BOHIN (continued)
Examination-in-chief by MR JOHNSON

MR JOHNSON: Good morning, Dr Bohin. Is it all right if the
witness --

MR JUSTICE GOSS: Yes, please, do sit down. Obviously you're
still subject to the oath that you took.

MR JOHNSON: Dr Bohin, I would like, if we may, please, to move
to the case of [Baby B]. You have written one substantive report
on [Baby B] dated 17 March 2019; is that correct?

BOHIN: That's correct.

JOHNSON: Thank you.

Now, turning to -- first to the material that you were asked to look at. It's set out in paragraph 4.3 of your report, which is about halfway down or two-thirds of the way down the second page.

BOHIN: Yes.

JOHNSON: Were you given a copy of [Baby B]'s medical notes from the Countess of Chester Hospital?

BOHIN: Yes.

JOHNSON: Copies of the X-rays or radiographs?

BOHIN: Yes.

JOHNSON: And the witness statement from Dr Evans of 31 May 2018?

BOHIN: Yes.

JOHNSON: So in other words not his initial sift report, but his final, in inverted commas, report?

BOHIN: That's right.

JOHNSON: And were you also given a couple of photographs?

BOHIN: Yes, I was.

JOHNSON: And can you just explain, just remind us, please, what your role was so far as your instructions are concerned?

BOHIN: I was asked to peer review the report of Dr Dewi Evans and I was asked to review case notes of babies, in this particular case [Baby B], and to ascertain, on the evidence contained within the case notes and the information I was given, if I could find a cause for her collapse.

JOHNSON: Thank you. I'm not going to ask you to explain this every time you return to give evidence, and I'm well aware you have told the jury once, but just to remind the jury please on this occasion, what is peer review?

BOHIN: It's reviewing what someone else has done before you.

So it means looking at what others have done. It doesn't mean completely reviewing things from the start, but of course you do have to, in a clinical context, look at the case notes. But it means to review what someone else has done before you, a colleague, someone of the same sort of standing as yourself.

JOHNSON: When you say "in a clinical context", most of us use clinical in the sense of cold-blooded. What does clinical mean in the way you use the word?

BOHIN: Forgive me. Clinical, my term, is something that is in a medical context, so looking at medical notes.

JOHNSON: Yes, all right.

You set out in your report [Baby B]'s background which in essence is a summary of her life in hospital; is that right?

BOHIN: It is. Dr Evans had already written in great detail about her -- the early part of her life, so I just wrote what I thought were the salient points.

JOHNSON: Yes. And you record as a matter of fact the circumstances surrounding [Baby B]'s collapse. I'm now looking at paragraph 6.8 of your report.

BOHIN: Yes.

JOHNSON: That was, just to remind the jury, at 00.33 in the early hours of 10 June.

BOHIN: Yes.

JOHNSON: You then set out further details from the notes, and you came to an opinion at section 7 of your report which is at the bottom of page 5 of 7; is that right?

BOHIN: That's correct.

JOHNSON: So far as the time between [Baby B]'s birth, just after 8 pm on the 7th, and her collapse at just after 00.30 on 10 June, what conclusion from the notes did you reach as to her clinical state?

BOHIN: At birth she was clearly compromised and needed some help to establish normal -- a normal breathing pattern.

So she did need resuscitation at birth, and that would not be particularly unusual for a baby born at just after 31 weeks' gestation. But she responded very well to appropriate resuscitation provided by the medical and nursing team.

Thereafter she needed some respiratory support in the form of either BiPAP or CPAP, and again that would be the kind of respiratory support that you would expect a baby of that gestation to need.

But she stabilised very quickly after the initial resuscitation and was stable -- she was initially put on BiPAP, which is a specific form of CPAP, it is CPAP with a little bit of extra support, but she did very well on that. So she was then converted to CPAP, but was in air, and her blood gases were normal. She was in air.

She was very, very stable. Stable enough, in fact, to have times off CPAP, having skin-to-skin contact with her mother, and also for feeds to be started.

JOHNSON: In terms of the time -- the duration of the time that she was off CPAP, I think we've made reference to the duration of that in paragraph 7.1. On your assessment of the notes that were being made at the time, what -- was this a case of [Baby B] coming off for 10 minutes or what?

BOHIN: In this particular case [Baby B] was taken off for 2 hours. There isn't a standard protocol for how long you would take babies off CPAP, but as Dr Evans said yesterday, in order to keep babies on CPAP, they have a kind of special hat and a bonnet with Velcro ties. So to get them on and off CPAP is a bit of an ordeal. So you wouldn't take a baby off for just 10 minutes, because that would be too much handling for little benefit.

In this case [Baby B] was taken on, on the first occasion, for 2 hours, and then was electively put back on. That was the plan.

On the second time she was out for almost two and a half hours, and was having skin-to-skin contact with her mother. So that meant that she was very close to her mother's chest, and we know that has very positive benefits for both babies and mums.

JOHNSON: Is there any suggestion from the notes that that extended time without any breathing support at all was causing any distress to [Baby B] or compromising her in any way?

BOHIN: Certainly in the first 2 hours off, there was -- there were no hiccups at all. There was no change in her observations, so saturations, her respiratory range, her heart rate stayed the same.

The observations also remained the same during her second period off, but there is a documentation in the notes that she was put back on CPAP because of increased work of breathing, although

that isn't reflected in the observation chart because the observation chart just tells you the number of breaths per minute, it doesn't tell you if you're struggling to breathe or if the baby is compromised.

But there was nothing in the observation chart would suggest that she was compromised in any way.

JOHNSON: Was it necessary to use oxygen that second time --

BOHIN: Oh, no, no.

JOHNSON: Thank you. Having read the notes and having listened to the evidence surrounding the circumstances in which [Baby B] collapsed, are those circumstances that you have heard normal, in inverted commas?

BOHIN: No. They are very concerning because babies of 31 weeks' gestation, although premature, you might anticipate that they might have some problems.

[Baby B] -- the only problem [Baby B] had was that she was requiring some CPAP support. But despite being on CPAP, everything else was completely normal and stable, and so to suddenly collapse is not something that you see in babies of that gestation.

If they have a deterioration, you normally get some prior warning: maybe they don't tolerate their feed, they have some desaturations, they have some bradycardias, maybe. She had nothing that would suggest that she was suddenly going to collapse in this way.

JOHNSON: So no warning signs at all.

There was, before the collapse at 00.33, a circumstance in which her nasal prongs became dislodged. Have you reached any views at all about that, that particular event?

BOHIN: Nasal prongs can be misplaced. They usually fit quite snugly in the nose to form a seal in order to give the CPAP, but they can be misplaced. If that's the case, they're probably quite uncomfortable because they are not sitting in the nostril, they are sitting at the side of the nose and that can be uncomfortable.

If, over a period of time, they're left off, the baby may desaturate. But in this case it was noted, the prongs were replaced, and everything went back to normal.

JOHNSON: And so far as you were concerned, in your professional opinion, is there any connection between the displacement of the nasal prongs and the desaturation at 00.33?

BOHIN: No. None at all. There were no -- that was a kind of a mechanical problem. It was rectified and [Baby B] was fine. There were no other warning signs that would herald an imminent collapse.

JOHNSON: And so considering the circumstances of the collapse, the nature of the collapse and the physical descriptions that you have heard given by the witnesses who saw the collapse or its immediate aftermath, what conclusion have you reached as to why [Baby B] collapsed?

BOHIN: Well, on reviewing the evidence, I had to formulate what could possibly have caused this, and make a differential diagnosis, so a list of things that could possibly cause a baby to collapse, and then look at the evidence in front of me to see if any of those fitted with the way that [Baby B] collapsed, what she looked like when she collapsed, and how she subsequently recovered.

And so the usual things that you would expect in a premature baby, things like infection, could be discounted. Cardiac arrhythmias could be discounted because they just don't simply present in at that way.

So I was left with the conclusion that this was an air embolus.

JOHNSON: Was it simply that you had discounted all the other possibilities that led you to that conclusion or was there evidence that in itself in your view supported that as a conclusion?

BOHIN: Well, I think partly it was a diagnosis of exclusion, but the important thing was the very florid descriptions of this rash and the skin -- not a rash, the skin changes that were present during this.

Everybody that's commented on it has said how -- (a) how florid it was, how different it was from anything they'd ever seen before, and I know from researching the literature that this is found in cases of air embolus.

So it was the rash plus -- and the excluding other potential causes because it just didn't fit with any other known potential causes.

JOHNSON: In reaching your conclusion were you taking into account at all the circumstances surrounding the collapse of [Baby A]?

BOHIN: I looked at this case on its own -- on its own -- on its own merit. But, you know -- and that is what I did for all the cases I have looked at. I have looked at this case on its own merits.

MR JOHNSON: Thank you, Dr Bohin. Would you wait there, please, there will be some more questions for you.

BOHIN: Of course.

Cross-examination by MR MYERS

MR MYERS: Dr Bohin, I'm going to ask you some questions on behalf of Ms Letby.

Some general questions first and then we are going to have a look at what you say about the two children you have reviewed.

You began your professional career, in terms of special, as a paediatrician, didn't you?

BOHIN: No, I was a neonatologist.

MYERS: I'm not going to dispute you are a neonatologist. I wanted to confirm, if you began as a paediatrician and then became a neonatologist, that's all.

BOHIN: Junior doctor training, at that time, you trained as a neo -- trained as a paediatrician, but at that time you could also train and be accredited in neonatology, and I was accredited in paediatrics and neonatology, which was the only accreditation you could have at that time. Things are different now.

MYERS: And you've been a consult neonatologist during your professional life, haven't you?

BOHIN: Yes.

MYERS: And you'd agree, would you -- would you agree there's a difference between a neonatologist and a paediatrician?

BOHIN: Well, all paediatricians, as part of their training, then and now, have to undertake some neonatology as part of their training. So all junior doctors, whether they end up being a consultant paediatrician or a consultant neonatologist, have to undertake neonatal training at registrar level.

MYERS: Is there a difference between a consultant paediatrician and a consultant neonatologist?

BOHIN: There is in the neonatologist ends up working solely in neonatology, whereas a paediatrician may work in a smaller centre. Neonatologists work in tertiary centres and they don't work anywhere else.

Paediatricians work in smaller hospitals where they cover both neonates and paediatrics.

MYERS: Neonatology is a specialised discipline, isn't it?

BOHIN: Yes, it is.

MYERS: Focusing on the first 4 weeks of life?

BOHIN: Well, it focuses on longer than that, because babies who are born prematurely often stay in hospital for many months.

MYERS: But a particular focus is that period after birth, isn't it?

BOHIN: It's the time from -- the time a baby is born until the time the baby is ready to leave a neonatal unit.

MYERS: Do you think, in your professional expert opinion, Dr Bohin, that someone who spends a large part of their career working with neonates will recognise that subtle features can have a significant impact in their presentation?

BOHIN: Yes.

MYERS: Yes. It's not always obvious what's going to happen, is it?

BOHIN: Not to anyone.

MYERS: No. Because if it was, deteriorations that are sudden and unexpected wouldn't happen at all, would they?

BOHIN: Sorry, could you rephrase -- I don't understand the question, I'm sorry.

MYERS: If it was obvious when something was going to happen, deteriorations that are significant could be stopped, they wouldn't get that far at all, would they?

BOHIN: That's not quite true because the majority of the time in neonates, you do get subtle changes and warnings before a baby

collapses. It's very unusual for a baby to go from being well to a collapse requiring resuscitation.

MYERS: Those changes can be quite subtle?

BOHIN: Not always, no.

MYERS: They can be, can't they?

BOHIN: They can be.

MYERS: Yes. Do you agree that it's important for an expert, or someone who holds themselves out as an expert, to have current clinical experience?

BOHIN: Not necessarily, no, because I think what you can't do is dispel the experience that a person may have had over many years. As long as they are current in their thinking and their reading, clinical practice -- and they are up to speed with clinical practice, the fact they are not doing day-to-day hands-on work, I think, makes little difference at all. In fact, lots of tertiary centre consultants don't do very much hands-on work, but that doesn't mean they are not credible.

MYERS: Do you not regard it as important for an expert to be working in a current clinical environment in relation to a case that they are considering?

BOHIN: No.

MYERS: You don't think that's important?

BOHIN: I don't think it's crucial, no.

MYERS: No. Are you -- do you think it's important for experts to develop their knowledge in clinical practice on an ongoing basis?

BOHIN: That's the same thing, isn't it?

MYERS: I don't know. Do you think it's important? Sorry if it's a repetition?

BOHIN: No.

MYERS: Do you think experts can gain from colleague-to-colleague discussions about the babies they're dealing with day to day, and that enhances their knowledge?

BOHIN: That's absolutely part of clinical practice that you discuss with your colleagues problems that you may have

encountered during your working day. But you also get that same sort of information by networking, going to conferences, and networking in that way. You talk.

MYERS: What I asked you was: do you think it's important for them to have that type of contact with colleagues on the job, clinical contact?

BOHIN: It's advantageous, but it's not essential.

MYERS: It's advantageous.

You're aware that experts in court proceedings have particular duties, don't they?

BOHIN: Yes.

MYERS: Some of those are set out by the courts, that's correct, isn't it? And indeed the GMC, the General Medical Council, which regulates doctors who practise, that also has standards on expert evidence, doesn't it?

BOHIN: Yes, it does.

MYERS: You're familiar with those?

I just want to ask you about a section in those rules called:

"Acting as a witness in legal proceedings. Giving evidence as an expert witness."

You're familiar with that?

BOHIN: Yes.

MYERS: Right. And maybe just -- I don't know if we can access D7, please. Perhaps at this point for Dr Bohin and for the -- his Lordship and the lawyers.

Just go over, please, to paragraph 12. Can we enlarge paragraph 12 so Dr Bohin can see this. I'm going to read out what paragraph 12 says:

"You must only give expert testimony and opinions about issues that are within your professional competence or about which you have relevant knowledge including, for example, knowledge of the standards and nature of practice at the time of the incident or events that are the subject of the proceedings. If a particular question or issue falls outside your area of expertise, you should either refuse to answer or answer to the best of your

ability, but make it clear that you consider the matter to be outside your competence."

I only read the second bit for completeness.

Certainly the GMC, in their guidance to experts, regard it as important that someone holding themselves out as an expert should have knowledge of the standards and nature of practice at the time of the incident or events that are the subject of proceedings, don't they?

BOHIN: They do, but they don't say you need to be in clinical practice.

MYERS: They said you should have knowledge of the standards and the nature of the practice?

BOHIN: Yes.

MYERS: That is enhanced by being in practice at that time, isn't it?

BOHIN: No, not necessarily, and it doesn't say that.

MYERS: Someone who is not in clinical practice will not have the same knowledge of the standards and nature of practice as somebody who is, will they?

BOHIN: Yes, they certainly can.

MYERS: How is -- just assist us with this: if somebody has retired from clinical practice in 2009 and is dealing with events that happened in 2015, do you say they are in the same position as someone who is in practice at that time?

BOHIN: If they have kept themselves up to date with clinical practice, then yes, absolutely.

MYERS: If they have done that. In any event, your view, so I can make this quite clear, is that being in current clinical practice is no advantage to an expert; is that your view?

BOHIN: I didn't say that.

MYERS: Is it of advantage to an expert then, if you didn't say that?

BOHIN: It would be advantageous, but not essential, I think were the words I used.

MYERS: Advantageous. Why?

BOHIN: Because you are seeing day-to-day events, you're able to talk to peers. But you have still got a level of expertise that someone who is retired would have. They don't lose their expertise because they've retired.

MYERS: So earlier I asked you if being in day-to-day practice and talking to peers was something that was significant and you disregarded that, didn't you?

BOHIN: No, I didn't disregard it. I said it wasn't essential.

MYERS: And you accept in fact it is an advantage?

BOHIN: It could possibly be an advantage.

MR JUSTICE GOSS: I think, in fairness to the witness, she did say it was advantageous but not essential to have clinical contact with other practitioners at the time.

MR MYERS: I apologise if I (overspeaking) --

MR JUSTICE GOSS: I think that's what you've just repeated essentially, yes.

MR MYERS: Peer review is not the same as approaching this as a kind of cold review, is it, if you see what I mean?

I can explain what I mean if you don't.

BOHIN: If you could, I would be grateful.

MYERS: Yes, of course. In peer review, your focus is looking on the work in this case that Dr Evans, in particular, has done, isn't it?

BOHIN: It is.

MYERS: It's not a case that you were handed all the papers and asked, absent whatever anybody else thinks, what do you think? It's not what has happened with you, is it, Dr Bohin?

BOHIN: Yes, it is. I was asked to peer-review Dr Evans' reports, but it was made very clear to me that I should form my own opinion about whether I agreed with Dr Evans' views, whether I disagreed with Dr Evans' views, or if I found an alternative explanation for why the children whose cases I reviewed collapsed.

MYERS: We can take this off the screen, by the way, Mr Murphy.

Thank you, Dr Bohin, I didn't mean to be rude when I said that.

But by the time you came to look at the material in this case, you had a very clear idea what the allegations were, didn't you?

BOHIN: I knew that there was a problem with babies collapsing at the Countess of Chester Hospital.

MYERS: Was it 2019 when you became involved with this?

BOHIN: Yes.

MYERS: That's after the initial media reports and coverage of what had taken place at Chester, isn't it?

BOHIN: Yes.

MYERS: So you had an idea of what this was about, didn't you?

BOHIN: Yes.

MYERS: Right.

You were given Dr Evans' opinions via his reports, weren't you?

BOHIN: Yes.

MYERS: Naturally. Did you have contact with Dr Evans?

BOHIN: No.

MYERS: Have you had contact with him over the course of these proceedings?

BOHIN: The first contact I had with Dr Evans was in -- I have never met Dr Evans before. The first contact I had with Dr Evans was earlier this year, in 2022, when we were asked to discuss a case -- one of the cases in this trial to formulate -- just to have a discussion about it because we had a difference of opinion. So that was the first time, and that was a telephone conversation because obviously I live in Guernsey, Dr Evans lives in the UK. I had never met him before.

MYERS: Have any of your opinions been influenced by the mere fact of the allegation that is made rather than the facts relating to the case?

BOHIN: No, I took each case on its merits to try and -- looking at the evidence, to try and ascertain why each child collapsed.

MYERS: And are you able to give your opinions independently of Dr Evans, Dr Bohin?

BOHIN: Yes, of course.

MYERS: You've been provided with the witness statements in this case, haven't you?

BOHIN: Yes.

MYERS: And the medical exhibits?

BOHIN: Yes.

MYERS: And certainly, up to this point, you've been able to watch what has happened either in court or on the link, haven't you?

BOHIN: Yes, I have.

MYERS: Just before we turn to [Baby A], can I just ask, when, if ever, let me ask, did you last see a case of air embolus in your clinical practice?

BOHIN: I have seen one case of air embolus in my clinical practice, and it was when I was a senior registrar in Leicester. So I can't remember the exact year, but as I became a consultant in 1996, and was a locum for a year before that, it would have been before 1995.

I worked in Leicester in a cardiac hospital and at the time there was a national trial for a treatment called ECMO treatment, which is a kind of cardiopulmonary bypass. It's a very specialist treatment, only done in specialist hospitals, and there was a neonatal trial going on, a national neonatal trial going on, which happened to be centred in Leicester.

I was part of the team in that trial, and I happened to be working in a hospital, the cardiac hospital where the ECMO was taking place, and it's a recognised complication of ECMO, air embolus is a recognised complication of ECMO where you get air bubbles in the cardiac bypass circuit.

So I saw one case, and that would have been a neonate, but not a premature neonate, because you can't give ECMO to premature neonates.

It's so long ago. I remember that it was an absolute emergency, but I can't remember anything else about it. I just remember seeing air bubbles in the ECMO circuit, which was absolutely striking.

But that's the only case I have seen in my career.

MYERS: So in terms of the conclusions you reach as to air embolus in this case, that's based -- that's based upon what --

MR JUSTICE GOSS: Just pause. (Pause). Yes, thank you.

MR MYERS: That's based upon what you say you are left with when other alternatives have been excluded; is that correct?

BOHIN: Partly.

MYERS: Partly?

BOHIN: Partly. The other is the very florid explanation and description of the rash.

MYERS: Yes. The skin discolouration?

BOHIN: Yes.

MYERS: That's the other thing that's been influential in your views, is it?

BOHIN: It helped me to formulate my views once I had excluded other possible causes of a sudden collapse because it's so unusual.

MYERS: In terms of excluding causes, can I ask you -- I asked Dr Evans the same thing so let me just ask you: where there is a serious deterioration in the health of, in this case a baby, or a death, do you agree that may be due to an identified medical condition?

BOHIN: Yes.

MYERS: I am just talking about the process at this point, not any particular case. It may be due to a medical condition that is currently known but hasn't actually been identified?

BOHIN: I'm not sure I understand that.

MYERS: With, when I was asked that question by Dr Evans, I made reference to the fact there are other tests sometimes or other conditions that may not have been considered in a given case that may apply.

BOHIN: I think yesterday with Dr Evans you mentioned things like genetic and metabolic conditions. Usually in babies, if they have either of those, there's an indication that they may have

those either by maybe abnormal facial features or other things you find on testing.

So they don't usually come out of the blue either.

MYERS: Sadly, in the case of [Baby A], there wasn't a lot of time for testing, was there?

BOHIN: No, but there was nothing in his presentation that would make the doctors looking after him think that he had a genetic or a metabolic or another rare disorder.

MYERS: Well, whether there was or there wasn't, of course, we know something happened to him, didn't it?

BOHIN: Yes, it did.

MYERS: And you cannot exclude genetic causes from how someone presents in a 24-hour period just after birth, can you?

BOHIN: I don't know of any genetic condition that causes you to collapse and die within 24 hours of life.

MYERS: But if we want to, for instance, consider alternatives, or look for them, genetic screening to identify a metabolic disorder is something that can be done, isn't it?

BOHIN: Genetic screening doesn't identify a metabolic disorder; they're separate things.

MYERS: Well, let's start with genetic screening. That can be done, can't it?

BOHIN: You would only do a genetic screen if you felt the child possibly had a genetic condition. When babies come to a neonatal unit, they are not screened for every conceivable illness or abnormality. You are led by the clinical presentation of the child at the time.

So genetic testing would only be carried out if there was a clinical suspicion that a child had a genetic condition, which wasn't the case.

MYERS: But it may not always be obvious, might it, Dr Bohin, whether there was or there wasn't. I'm just asking about the process at the moment.

BOHIN: It may not be obvious, but I can't think why anyone would think of doing a genetic test.

MYERS: Maybe if they want to exclude all possible alternatives, they might do this then, mightn't they?

BOHIN: That's not the way clinical medicine works.

MYERS: We are talking about the forensic exercise in a courtroom when someone is on trial for these charges.

They might try to seek to exclude alternatives then, mightn't they?

BOHIN: Sorry, I don't understand the link, I'm sorry.

MYERS: If somebody wants to exclude possibilities, if somebody is going to consider other medical conditions that are maybe known but unidentified, one step that can be taken is genetic screening to see if there is any underlying condition that has not yet been identified. Do you agree?

BOHIN: No, because you would only do that if there were physical manifestations or clinical manifestations that led you to think that there may be a genetic reason for the underlying presentation. You wouldn't do it as an academic exercise.

MYERS: This is not an academic exercise.

BOHIN: No, I'm not --

MYERS: So your mind didn't turn to that in any event, Dr Bohin?

You say it wouldn't turn to that?

BOHIN: No.

MYERS: All right. It may be due to a medical condition that's currently unknown. It sounds something that is less likely, but it's possibility, isn't it?

BOHIN: It is.

MYERS: It may be due to a problem in the care that is given?

BOHIN: I don't think his collapse was related to a problem in the care that was given.

MYERS: Problems in the care that is given can be related to collapse in principle, can't they?

BOHIN: In principle, yes.

MYERS: Right. Of course it may be due to deliberate inflicted harm?

BOHIN: It could be, yes.

MYERS: Do you agree it's important not to start with an assumption, for instance, that there has been deliberate harm?

BOHIN: Absolutely.

MYERS: Do you agree it's important not to hurry to a conclusion of deliberate harm?

BOHIN: Absolutely.

MYERS: Do you agree the wrong approach would be to start out with a the theory there had been deliberate harm and then look for those things could support that?

BOHIN: Yes, I agree with that.

MYERS: Do you agree it would be appropriate in a case like this to make any further investigations or enquiries that could possibly be relevant if they point away from deliberate harm?

BOHIN: Yes.

MYERS: Turn to [Baby A], please.

Your evidence, just so I can confirm this is correct, is that before the time of the collapse [Baby A] was extremely -- extremely stable and not escalating in any problem; that is your evidence, yes?

BOHIN: Yes.

MYERS: Could you just help with us the difference between stable and extremely stable?

BOHIN: That's subjective. In my view there was nothing that gave cause for concern with [Baby A]. So I thought he was extremely stable.

MYERS: There had been some breathing problems from birth, haven't there?

BOHIN: As you would expect.

MYERS: Let's just look at some of the documents we've seen already, but as is right, you have the opportunity to comment on what questions are asked.

Could we look at the blood gas reading at 1125, which is S6, please, Mr Murphy. If we can enlarge the relevant area, please.

Do you agree, looking at that, Dr Bohin, that there is first of all an increase in lactate, looking across to the right-hand side?

BOHIN: Marginally.

MYERS: Yes. And at 2.6 and 2.7, it's above what is the level -- optimal level, isn't it? It's elevated?

BOHIN: In my hospital the upper limit of normal is 2.5. But in the countless Chester it's 2., so yes, it's a bit elevated. But you obviously have to take that in conjunction with the other parameters, not as a single reading.

MYERS: Well, let's look at the other parameters because we can see blood oxygen appears to be dropping. Whatever the merits of a heel prick test, it's dropping, isn't it?

BOHIN: You can't read anything at all into the PAO2, which is the oxygen level in a capillary sample, which is why in the left-hand side there are the letters A, V and C that are circled, so that it allows the person interpreting the test to know whether this is an arterial sample, where you can read something into the level of oxygen, or a capillary or venous sample where you cannot.

MYERS: So you can't read anything into it if it drops?

BOHIN: No.

MYERS: So it could actually be much higher than that?

BOHIN: Yes.

MYERS: So it could actually be much lower than that?

BOHIN: Yes.

MYERS: So what was done to check what the blood oxygen was?

BOHIN: You've got the other parameters and you've got a saturation monitor to the baby.

MYERS: So this is of no particular use at all, the PO2, if it's capillary?

BOHIN: No.

MYERS: Can we look at the observation chart at 1123. It's at S7, please.

We can just look at respirations first, please.

Look at respirations and heart rate together.

Whatever the reason that was given, the respiration rate is not stable, is it?

BOHIN: It's above the normal range.

MYERS: Well, it's not stable, is it?

BOHIN: It's -- I would say that it is stable, yes.

MYERS: And whatever the explanation, as we go through the afternoon, it's actually elevating, isn't it?

BOHIN: Which afternoon are you --

MYERS: The afternoon of the 8th going into the 9th. It's in fact the afternoon of the 8th. Can you see where that is, Dr Bohin?

BOHIN: Yes.

MYERS: That's elevating, isn't it? In fact, starting from before then, Mr Murphy. If we start from the morning in fact.

It's come down after birth but now it's coming up, isn't it?

BOHIN: So the range is between 60 and 80 breaths per minute which is outside the normal range, but this was a baby on CPAP, receiving respiratory support, but was in air.

So any respiratory underlying pathology could not have been significant because the baby (a) would not have been just on CPAP, would have needed more invasive ventilation, and would not have remained in air and would not have had saturations that were normal.

MYERS: When we look at that chart, I would just like to have an answer to the question if that's all right, that's escalating, isn't it?

BOHIN: Not in every case because at 15.00 it's back down to 60, and then it goes up to 80. And during that time there would have been interventions on [Baby A] which may have caused his respiratory rate to rise.

MYERS: So in your professional expert opinion, so I can be clear, you don't regard what we see from 7 o'clock in the morning to what we see shortly before the collapse as an elevation? You don't regard that as an elevating respiratory rate?

BOHIN: I don't regard it as a worrying sign, no.

MYERS: What's the point of having it in the yellow?

BOHIN: The point of having it in the yellow is to alert the staff looking after the baby that this is above what we could consider to be the normal range, but in conjunction with the other things that are being monitored, the team need to decide: is this worrying so that we need to intervene and maybe put the baby on to more invasive respiratory support, ie ventilate, or is it something that we are aware of but we will just watch because the other things that we're monitoring are all normal? And that was the case.

MYERS: So you say it's something that is an alert?

BOHIN: It's an alert for people to be aware of.

MYERS: And have you seen anything in the notes indicating that anyone took any action as a result of this being an alert?

BOHIN: People noted it.

MYERS: Well, they noted it.

BOHIN: But they didn't do anything.

MYERS: They didn't do anything?

BOHIN: They didn't do anything because nothing needed to be done --

MYERS: That rather prejudges the outcome, doesn't it, Dr Bohin?

BOHIN: Well, he was in air, so there was nothing that could be done. He didn't require ventilation, which would have been the next step. He wasn't having lots of desaturations, which would lead to ventilation. So the next step in terms of respiratory support would have been to ventilate him and he didn't require that.

MYERS: Can we just look at the bottom of the form for a moment, please. Can we just go right down to the bottom of the form. Just enlarge what it says at the bottom. Can you see?

"White: continue observations as per frequency.

First yellow: action and repeat. Second yellow: escalate to senior nurse practitioner or a doctor, [an SHO]."

That's what it says you should do?

BOHIN: Yes.

MYERS: Have you seen anything to indicate this was escalated to a senior nursing practitioner or a doctor?

BOHIN: At which time?

MYERS: Second yellow.

BOHIN: So the first yellow would have been -- so the respiratory rate being abnormal. The second yellow, I assume, means if the baby had an abnormal heart rate or abnormal temperature or something else that would require escalation.

MYERS: Right. Can we go back up then, please.

Let's look at the heart rate. Yesterday Dr Evans explained to us that this could all be due to the handling that was taking place of [Baby A] as the lines were being put in and the UVC. Do you remember him saying that?

BOHIN: Yes.

MYERS: And handling can cause discomfort to a neonate, can't it?

BOHIN: Yes.

MYERS: And it can cause stress?

BOHIN: Yes.

MYERS: And does that lie behind what he said about, "This is why respirations will go up", or what he suggested?

BOHIN: That's a matter for Dr Evans.

MYERS: Right. But in your view is that an accurate assessment of why they may go up?

BOHIN: Handling in a baby on respiratory support can make the respirations go up, yes.

MYERS: If respirations go up because of that type of stress or distress, ordinarily you would expect that to be accompanied by a rise in heart rate, wouldn't you?

BOHIN: It may do but it doesn't have to. It's not a given.

MYERS: Wouldn't you expect it?

BOHIN: Not necessarily, no.

MYERS: If a baby is well and things are functioning as they should, wouldn't you expect heart rate and respiratory rate, to some extent, track one another in terms of they're proportionate in the profile?

BOHIN: No.

MYERS: If we just look at -- if we go -- we're familiar with this. Can we just look at the [Baby B] timeline, please.

It's 1624, slide 19. 1623, if we could get there first, but I'm not sure there's a slide, Mr Murphy. 1623.

Thank you.

Now, just look at the top half. It's respirations and heart rate that I'm asking us to look at. This is [Baby B] over the same period.

BOHIN: Yes.

MYERS: Can you see heart rate and respirations largely are proportionate to one another, they almost track one another? I'm not talking about whether they are in white or yellow. I'm talking about the profiles.

BOHIN: I don't think they necessarily do, no.

MYERS: Well, we are all looking at it. We can hear your evidence, Dr Bohin.

Can we have a look at the next page, page 1624, which is on the timeline at slide 19.

Again, it may be more pronounced in heart rate, but do you agree that, to a large extent, wherever they are with regard to white and yellow, they're following similar profiles?

BOHIN: No, they are not at all because the heart rate is going up, but the respiratory rate doesn't go up correspondingly there at all.

MYERS: Then if we go back, please, to 1123, slide 7 -- sorry on the [Baby A] profile. Thank you. We can just enlarge again, please.

We can see there an increase, going up in the respirations, not reflected by any similar increase in heart rate between 8 and 2 on 8 June.

(Pause)

BOHIN: Could you repeat that so -- on 8 June between which times, sorry?

MYERS: The same time we were looking at earlier, Dr Bohin, between 8 o'clock, an almost twelve-hour period between 8 in the morning --

BOHIN: Yes. The respiratory rate is up and down and the heart rate is up and down and they don't necessarily correspond.

MYERS: The respiratory rate elevates, doesn't it, from about 8 o'clock to where we get to just before the collapse, doesn't it?

BOHIN: It also goes down at 15.00 and 18.00 --

MYERS: Right. And the overall direction of travel is up, isn't it?

BOHIN: No.

MYERS: All right. You say no?

BOHIN: No.

MYERS: And the heart rate appears to be, if we look across the whole band, relatively steady, doesn't it?

BOHIN: That goes up and down as well, but not in conjunction with any change in respiratory rate.

MYERS: We know that -- would you agree this is significant tachypnoea?

BOHIN: I don't think it's significant tachypnoea, no.

MYERS: The yellow, you have said, indicates that someone should take some sort of action; is that correct?

BOHIN: No, I said that you needed to note it. I didn't say you needed to necessarily take any action. I said you needed to note it and look at the other parameters that were being monitored.

MYERS: In any event, your view is that there's nothing there that shows an elevating respiration; is that correct?

BOHIN: Yes.

MYERS: And your view is that the heart rate there tracks what the respirations are doing; is that your view?

BOHIN: No, the heart rate is doing its own thing.

MYERS: It's own thing? Right.

BOHIN: It's up and down, but it's not rising or falling in conjunction with the breathing rate.

MYERS: And you say there's nothing to worry about if the heart rate is doing something different or not in conjunction with the breathing; that's your evidence?

BOHIN: Yes.

MYERS: And you say that doesn't indicate any issue, for instance, with the heart?

BOHIN: You can't tell anything about the heart from this. All you can tell is how fast the heart is beating. You can't tell whether it is functioning well or otherwise.

All you can tell is the number of heart beats per minute from this.

MYERS: From the lines -- let's move to the lines that are were inserted into [Baby A]. There were a number of efforts to insert a UVC. There were two efforts, weren't there, and they both failed, and the attempt to insert a long line failed, didn't it?

BOHIN: Yes.

MYERS: In fact the long line was too close to the heart, wasn't it, when it was inserted?

BOHIN: It wasn't too close.

MYERS: Was it in a good position?

BOHIN: It wasn't in an optimal position.

MYERS: It wasn't in an optimal position? Right.

BOHIN: But it was in a usable position.

MYERS: We've heard Dr Harkness say that he and his colleagues were all of the view this was in a perfectly fine position, ideal really.

BOHIN: It was a safe position.

MYERS: Did you hear him say that?

BOHIN: Yes, I did.

MYERS: Do you agree with that?

BOHIN: I do.

MYERS: You do.

If we just look at your report, please, for 29 November 2020 for [Baby A], paragraph 5.5 -- do you have it there?

BOHIN: Yes, I do.

MYERS: What you said was:

"The X-ray showed the long lines have crossed the midline. It was not within the heart but it was not in the best position."

That's what you said when you wrote this report:

"It was not in the best position on 29 November 2020."

What's changed to the position since you wrote this report?

BOHIN: It isn't in the ideal position, but it's in a safe position.

MYERS: So -- what do you mean, not in the best position? You agree it wasn't in the best position? Do you stand by that?

BOHIN: It wasn't in that the best position, but it was in a safe position. You've got no control over where the line ends up, but it was not in a position that would be dangerous to use, and I think that's the important thing. It was in a position that was not going to cause any problem with the heart, because it wasn't in the heart, so therefore it was safe to use.

MYERS: If it is not going to cause a problem, how can you say it's not in the best position? Why not just say it's fine?

BOHIN: Because ideally you would want it to be in the -- depending on whether you've put the line in the leg or the arm, you would want it to be in a big vessel approaching the heart. This had actually gone to the other side, into a big vessel on the other side, but it wasn't in the heart, which is what you're worried about.

You're worried that the tip of the line may sit in the heart, and that is not a safe position.

So ideally you wouldn't want to cross the midline.

It did, but it was perfectly safe to use.

MYERS: So when you said, back in November 2020, that it was not in the best position, can I be clear, do you agree that there is any potential risk with it in the position it was in?

BOHIN: No, there wasn't any risk.

MYERS: And therefore why was there a need to pull back? So you can just explain, you say there was a need to pull back.

BOHIN: Dr Harkness decided he would do that, not me, and he decided that -- and I think in his testimony he said actually in retrospect he realises now that it was actually in a safe position but at the time decided to pull it back.

MYERS: Can I just ask this, before I move from this point. He described how he and his colleagues gathered the day afterwards and decided that there was absolutely nothing wrong about the line, it was well placed. Were they wrong to form that view?

BOHIN: No.

MYERS: So even though you say it is not in the best position, it's all right for them to form that view?

BOHIN: Yes.

MYERS: Right. We know he planned to pull it away, don't we?

BOHIN: Yes.

MYERS: But it remained there for 2 hours before it was used to put dextrose down, didn't it?

BOHIN: Yes.

MYERS: And it remained there because he wasn't able, because of the demands on the unit, to get there in time to pull it away?

BOHIN: That's right.

MYERS: And is that acceptable, that sort of delay with something like this? Do you regard that as acceptable?

BOHIN: I think it is inevitable in a busy neonatal unit.

MYERS: We hear -- we've heard already about people saying it's inevitable in a neonatal unit. Is it acceptable?

Inevitable is different, Dr Bohin.

BOHIN: It wasn't in a place that was dangerous. So I think the doctor on duty needs to prioritise his demands and anything that was dangerous, he would do -- he or she would do immediately.

This was not dangerous and there may have been other pressing concerns for him at the time. For instance, if he was called to a resuscitation or something that was life-threatening, that clearly took priority over withdrawing a long line, which in the event didn't actually need to be withdrawn.

So he would have to make his own decision about the priority of his workload at the time, and sometimes things fall down the list because emergencies take priority.

MYERS: Yes, but that doesn't make it right that they're not dealt with, does it?

BOHIN: Not everything can be done immediately. That's not the way that medicine works.

MYERS: No, but we are looking here at the consequences of the situation we're dealing with. Right?

Now, there's a possibility, of course, having been put in, a long line can move its position, isn't there, to some degree?

BOHIN: There is if they are in for a long time. Sometimes long lines stay in for 2 weeks or more, and they are taped in, in a particular way in the anticipation that they won't move, but we know that they can move.

MYERS: What would be the best way of knowing whether this long line had moved from what was, in your words, not the best position to a worse position? How would we know that for sure?

BOHIN: We would -- you would X-ray it. But in this case the tip of the long line had crossed the midline and what usually happens when -- where we're concerned about where the tip of the long line is, is that the long line actually advances in.

So it may start off in a great vessel, but over time actually goes in further into the heart, and that can cause a problem.

This long line had crossed the midline. It wasn't in that the heart. So if it went in further, it wouldn't have approached the heart, it would have just stayed in a vessel.

MYERS: Well, we don't know where the tip of the long line ended up, do we?

BOHIN: Well, in that very short spice of time, in a day, it wouldn't have moved. The recommendations are not to X-ray the tips of long lines every day. The recommendation -- there are recommendations about the care of long lines. There's a BAPM recommendation about the care of long lines because of the risk of cardiac tamponade. If the tip touches the heart, it can go through the heart muscle and cause a problem. So there are guidelines about that.

But the recommendations are not to X-ray the tip of a long line every day.

MYERS: Are you in any way, Dr Bohin, trying to explain away risks that could have contributed to collapse? Are you doing that in any way in your evidence?

BOHIN: No.

MYERS: What about going without fluid for 4 hours for [Baby A]?

Is that okay?

BOHIN: No, it's not okay.

MYERS: Why is it not okay?

BOHIN: Going without fluid for a short period of time would have no consequence, but of course [Baby A] wasn't being fed to any great degree. He was having some milk, but 1ml every other hour. So his total fluid volume was around 4ml per hour, and the 1ml every 2 hours of milk was being given as extra to requirements.

So not to have fluids for that period of time would -- would not be optimal care.

MYERS: Was it okay for him not to be receiving the blood sugar - - sorry, the glucose that he was meant to be receiving over that period of time?

BOHIN: That was the fluid. So yes, to not have fluid was not -- would not be considered to be great care. But again, he had no intravenous access and the doctors on duty had to prioritise what was the most important thing to do and what was not so important.

MYERS: But in saying that, you are talking about the reason for the risk rather than the risk, aren't you?

BOHIN: It's not ideal not to have intravenous fluids for 4 hours.

MYERS: And however that ends up, that can create conditions in which there could be a deterioration, doesn't it?

BOHIN: It wouldn't cause a sudden collapse in the way that [Baby A] collapsed, no.

MYERS: Well, first of all, what I asked you is: that creates conditions in which it's possible he could have a deterioration, isn't it?

BOHIN: The only deterioration he could possibly have would be to drop his blood sugar.

MYERS: And this is at a time we know with elevated respirations; yes?

BOHIN: He has got elevated respirations but I'm not sure the two are linked.

MYERS: You're not sure they're linked?

BOHIN: I don't think they are linked.

MYERS: I have asked you about his heart rate. The lactate has gone up. Is that consistent with elevated respirations?

BOHIN: On its own, no.

MYERS: Okay. In any event, would you agree the overall picture is suboptimal at that point --

BOHIN: No.

MYERS: -- given what we have with fluids, with respirations, lactate and not moving or repositioning the long line?

BOHIN: No, I don't agree with that.

MYERS: Right. Well, you certainly agree that the fluids are suboptimal, don't you?

BOHIN: Yes.

MYERS: All right. Then that makes the overall position suboptimal then, doesn't it?

BOHIN: No, it means that in the ideal world you would give fluids. I'm not linking that to the lactate, which I don't think is far outside the normal range, and in the context of everything else, I wouldn't pay any attention to it. The respiratory rate I think I have already explained.

MYERS: These are circumstances in which it is quite conceivable a baby like [Baby A], like this, could deteriorate dramatically because of the condition that he's in. Do you agree or disagree with that?

BOHIN: I disagree with that.

MYERS: You prefer the explanation of air embolus, don't you?

BOHIN: Yes.

MYERS: Right.

Discolouration is a key feature you rely upon in that, isn't it?

BOHIN: Yes.

MYERS: A key feature. Of course, that depends, first of all, on the quality of the descriptions that we're given, doesn't it?

BOHIN: Yes.

MYERS: Do you agree, Dr Bohin, there are many reasons for discolouration in infants?

BOHIN: Yes.

MYERS: And skin discolouration in and of itself is rarely specific to a particular -- I'm not talking about a meningococcal infection. In general, the mere fact of a skin change is not going to be specific to a particular condition?

BOHIN: Do you mean a colour range or a rash?

MYERS: A colour change.

BOHIN: No, a colour change is not specific.

MR JUSTICE GOSS: Perhaps you could just -- what is the difference between a colour change and a rash?

BOHIN: A rash is clearly a change in the skin. A colour change is literally that, a colour that goes from, if your skin is nice and pink, to very pale or blue or mottled. That would be a colour change.

A rash is something that all of us would know, if you've ever had chickenpox or you've had acne. There are lots of other rashes. Meningococcus is the one that people get concerned about, but that would be a rash, and that's not necessarily associated with colour change.

So it's a change in things in the skin. It may be pimples, it may be tiny little pinprick blotches like you get in meningococcal disease, but in itself it may not be associated with a change in the overall colour of the skin.

MR MYERS: Thank you, my Lord.

In any event, the discolouration, Dr Bohin, what we have -- whatever the descriptions are from the time, it's matter for everyone to assess as measured against what is described in the literature we've already talked about --

BOHIN: Yes.

MYERS: -- which is the article by Lee and Tanswell? Thank you.

I'd like to ask you about something else, please, and this is to do with where there may be an air embolus and how that may arise.

Do you agree that air embolus is a known risk from venous catheters?

BOHIN: It's been reported in the adult literature that it's a risk --

MYERS: Yes.

BOHIN: -- but I have never known it to be a risk in neonatal practice, and neither has there ever been an alert out to the neonatal clinical teams that this is something that you should be aware of.

MYERS: That doesn't mean it cannot happen, of course, in medicine, does it?

BOHIN: It does not mean that it can't happen, but if things happen commonly, like with the long line that causes tamponade in the heart, once there is a critical number of those, usually an alert would be generated nationwide to say this is a particular problem, and that hasn't happened with the insertion of central lines or UVCs in neonates.

MYERS: As much as you have explored with the prosecution your differential diagnoses, I am exploring a different thing here, so help us with this, please.

Do you agree that air can enter through the veins or into the venous system through a long line or a central catheter -- central venous catheter?

BOHIN: Yes.

MYERS: And as it happens, [Baby A] had a central venous catheter and a long line, didn't he?

BOHIN: Yes.

MYERS: And there have been some problem, as it happens, with positioning those during the day, hadn't there?

BOHIN: Yes.

MYERS: As to the prevalence of it, do you agree that what's called negative venous pressure can cause air to go in through a long line?

BOHIN: That's reported in the adult literature. I have seen no reports of it in the neonatal literature.

MYERS: No, all right?

BOHIN: Because the lumens of the lines are very, very small.

MYERS: But in any event let's talk about the mechanism of it.

You've actually provided a paper, haven't you, which describes this, a paper by Gordy and others, haven't you?

BOHIN: Yes.

MYERS: Just so there's no mystery, I'm going to go to the passage that deals with it, so everyone can know what it was you

provided. It's at page J52039. It will appear on the screen in a minute and we can see it.

Actually, if you look at the top, it's an article in the International Journal of Critical Illness and Injury Science. Thank you, Mr Murphy.

It's an article on vascular air embolism, Gordy and Rowell, but I'm going to go straight to the bit that deals with what we are looking at here.

If we go further down this page, Mr Murphy, down to aetiology, I'm going to read about the first seven lines. It says:

"Air embolism is a rare but potentially fatal occurrence and may result from a variety of procedures and clinical scenarios. It can occur in either the venous or arterial system depending on where the air enters the systemic circulation. Venous air embolism occurs when gas enters a venous structure..."

Venous, I'm sure we all follow, means to do with the veins, doesn't it?

BOHIN: Yes.

MYERS: "... and travels through the right heart to the pulmonary circulation."

And then this:

"Conditions for the entry of gas into the venous system are the access of veins during the presence of negative pressure in these vessels. This is most commonly associated with central venous catheterisation [like a UVC] as the potential for negative pressure exists in the thoracic vessels due to respiration. It can, however, occur in numerous clinical scenarios."

Pause there, just go back to that. It's there if anyone wants to go to anything else.

When it says:

"This is most commonly associated with central venous catheterisation, as the potential for negative pressure exists in the thoracic vessels due to respiration."

That means as, for instance, the patient breathes in a pressure gradient is created with the outside --

BOHIN: Yes.

MYERS: -- and air can be drawn in?

BOHIN: Yes.

MYERS: That's what they are describing, isn't it?

BOHIN: Yes.

MYERS: Yesterday when I asked Dr Evans about this he didn't accept that could happen and said he had heard of it but had problems trying to work out what it means. Do you remember him saying that?

BOHIN: Yes.

MYERS: You accept, Dr Bohin, it is something that can happen, don't you?

BOHIN: I have never seen it happen in neonates. Of course it can happen.

MYERS: Yes.

BOHIN: This is a -- this is a paper looking at adult practice.

MYERS: I am just seeing what you have to say on the same topic that I questioned Dr Evans yesterday. He said he didn't accept that it can happen. You agree that it can happen?

BOHIN: Yes.

MYERS: He said he had heard of it but had problems trying to work out what it means. You have no problem trying to work out what it means, do you?

BOHIN: No.

MYERS: Just on the subject of peer review and weighing this up, does it surprise you that for someone who is said to be or says they are an expert in neonatology, they do not accept this could happen? Does that surprise you, they say they do not accept this could happen?

BOHIN: No, we all have differences in opinion. So no, it doesn't surprise me.

MYERS: So he may well choose to say it doesn't happen?

BOHIN: Yes. He did.

MYERS: Does it surprise you he has problems trying to work out what it means?

BOHIN: No.

MYERS: The idea of a negative pressure gradient is rather like the way something is sucked through by suction or a siphon, isn't it, drawn through?

BOHIN: Yes.

MYERS: You agreed in your experience, in your view, that's not a difficult concept to follow, is it?

BOHIN: It isn't for me.

MYERS: No, it isn't for you. All right.

So, just so we can understand -- and take that down, thank you, Mr Murphy -- evidence of air embolus in this case is based primarily upon no other apparent explanation that you can identify; that's what you say?

BOHIN: Yes.

MYERS: You disregard any of the clinical features we have looked at in the course of these questions as being relevant to the collapse, do you?

BOHIN: Yes.

MYERS: And your view is there's nothing else that could have been done to look at any other line of enquiry?

BOHIN: Yes.

MYERS: You rely upon discolouration?

BOHIN: Yes.

MYERS: Do you agree that even taking that, nothing can establish conclusively, as it happens, that there was an air embolus in this case?

BOHIN: We have got post-mortem air that was shown on the radiographs.

MYERS: That's not something you say you rely on. I can come to that if you like.

BOHIN: No, I'm relying on the clinical picture in front of me.

MYERS: And we have heard a number of explanations for air in the radiograph, haven't we?

BOHIN: Yes.

MYERS: We have heard that that could be either because of air entering because of an air embolus, haven't we?

BOHIN: Yes.

MYERS: And that could be by accident or on purpose, couldn't it?

BOHIN: Yes.

MYERS: We have heard that that could happen as a result of a post-mortem gas developing?

BOHIN: Yes.

MYERS: And we have also heard that resuscitation, CPR, can lead to air being liberated and that can appear in this way?

BOHIN: Yes.

MYERS: All right.

I'm going to turn to [Baby B], please. I just want to ask this: what you said about your -- Dr Evans is entitled to his view on negative thoracic pressure.

Forgive me for asking, but you are not saying that as some sort of excuse for Dr Evans, are you, for him disputing that with me? Can I just ask, are you making excuse for him?

BOHIN: Absolutely not. That's a matter for Dr Evans.

MYERS: Does it not surprise you at least that someone who holds himself out as an expert in this does not recognise that mechanism even?

BOHIN: No.

MYERS: [Baby B]. We know with [Baby B] that the doctors were trying to place long lines into her at the same time as they are doing that with [Baby A], weren't they?

BOHIN: Not at the same time because it was the same doctor.

MYERS: Well, over the same time period, Dr Bohin.

BOHIN: Yes.

MYERS: The same doctor with both of them. In fact, is it right, that there were attempts by Doctors MacCarrick and Ogden to place a line into [Baby B] on 8 June?

BOHIN: An umbilical venous catheter.

MYERS: So it's the same procedure as with [Baby A], as it happens?

BOHIN: Yes.

MYERS: Were they successful?

BOHIN: No.

MYERS: No. Is that standard, four attempts across two children and none of them successful? Is that standard?

BOHIN: On occasion, yes.

MYERS: Is that optimal?

BOHIN: No, you would like to get it in first time.

MYERS: Leaving them like that, for instance, for any period, would that be suboptimal?

BOHIN: Not if they have got other means of receiving fluids, if there's another cannula, no.

MYERS: We know in [Baby A]'s case he wasn't receiving fluids for some period of time.

Can we just look at 1353, please. These references may not be on the iPads, ladies and gentlemen, but 1353.

I wonder if we could enlarge that, please.

If we look at the entry at 18.58 by Mel Taylor, who is the nurse who was also looking after [Baby A] in this period, 8 June 2015, we can see it says -- can you see it in the centre? Can we perhaps highlight that?

BOHIN: Yes.

MYERS: By this point we have two babies in nursery 1, both with venous catheters in place, don't we, with one nurse looking after both of them?

BOHIN: Yes.

MYERS: Right:

"UVC position in wrong place. Removed. Cannula tissue. SHO informed. Tried to cannulate as registrar busy on ward 30. Failed attempts. No fluids running for a couple of hours."

You can see some other matters recorded there. So it turns out that in [Baby B]'s case there's no fluids running for a period of time as well, doesn't it?

BOHIN: Yes.

MYERS: When I asked you whether it would be suboptimal, you said that depends on whether fluids could or couldn't run. So this has now become suboptimal, hasn't it?

BOHIN: Yes.

MYERS: And one of the reasons it is suboptimal is because the doctors required are elsewhere or the doctor is elsewhere.

BOHIN: Well, there is a doctor there --

MYERS: Yes.

BOHIN: -- an SHO who has tried but failed. So he or she has escalated and asked for a more senior doctor to come in and make an attempt, which is completely standard.

MYERS: Yes, and it takes a little while for that to happen, doesn't it?

BOHIN: Not always, but it did in this case because presumably the doctor was busy.

MYERS: Well, let's have a look, if we could, in the notes for -- at 1274. Again, it's not on the iPads, but let's just have a look at 1274. In fact it may be at S1. I'm just checking.

MR JUSTICE GOSS: 1353 was on the presentation but you have to go into the additional documents.

MR MYERS: I apologise. They have been added, I think.

MR JUSTICE GOSS: It's all right. It's no problem.

MR MYERS: Thank you, my Lord.

If we just look down here at 8/6/15, at the centre of the page, please, just enlarge that. It's 1274.

Now, we have heard that Dr Harkness was in the process, it seems, of attempting to fit a long line to [Baby B] at some point that evening. Do you recall hearing that evidence? He was in that nursery at some point trying to do that?

BOHIN: Yes.

MYERS: We are looking at a point here after -- this is 10.00 -- after [Baby A]'s collapse, aren't we? Can you see 8 June?

BOHIN: Is that 19.00?

MYERS: It may be. I can't quite tell. Do you think that's 19.00? Right, thank you.

19.00. It says:

"Inserted on fifth attempt by registrar Dr Harkness."

All right? So that would appear to be some period of hours before that was inserted, doesn't it, if that is when it first went in?

BOHIN: At 19.00 when the fluids had stopped?

MYERS: Yes.

BOHIN: Is that what you're suggesting?

MYERS: Yes.

BOHIN: Yes.

MYERS: Fluids had stopped, it seems, by whatever time was in the last note, early afternoon.

BOHIN: Four o'clock-ish, I think, something like that.

MYERS: So this is an even longer period than [Baby A] had been without fluids, isn't it?

BOHIN: Yes.

MYERS: Help with us this: is five attempts -- we've heard about the effect of handling on a baby. Is five attempts to insert a long line standard?

BOHIN: As I said before, you hope to get it in on the first attempt. The fact is you need to get a long line in and so you attempt as many times as you need to. If you are the most senior person on duty, it is down to you, and that may need five attempts.

MYERS: Is there a number which it's considered acceptable to have done it by?

BOHIN: Well, one.

MYERS: All right.

BOHIN: You would like to get it in the first attempt. Five attempts is -- that's what it took him. It's incredibly difficult to site these. It's incredibly difficult.

They're about a millimetre wide and the veins are very, very small. And so sometimes you can very easily get the needle and the tip into the vein, but it just won't thread for you, and, as we have heard, the -- you have to be able to get it so that it's in an optimal position, and if it won't thread, and you spend a long time fiddling around trying to get it to thread up, you're very aware that you are handling this baby.

So sometimes it's easier to take the needle out and try again in a different vein.

MYERS: And the handling of the baby can cause stress of its own; is that correct?

BOHIN: The handling of the baby can cause stress.

MYERS: And then having been put in, it seems, by about 19.00, if that's right, can we just look, please, at 1274 -- 1275. If we just scroll down, please. Just down to the bottom of that page.

Just enlarge that. In fact we may have to go back.

But this says X-ray -- this is following on from what we've looked at. In fact, so we can follow this through, Mr Murphy, can we just look at the bottom of the page we have just been looking at, please. Thank you.

This is at the bottom of page 1274.

If we just enlarge that, please, and then we can go on to the next one.

So this is the next entry. It says 11.00, but I'm going to suggest that may well be 23.00.

BOHIN: 23.00.

MYERS: Dr Lambie:

"Asked to withdraw long line by 3 centimetres.

Sterile. [Whatever that is]. Working well.

[Something] use."

BOHIN: "Okay to use."

MYERS: Can we look at the next page then to see what went into the records there?

So there's been an X-ray some time just after 10 o'clock, and the line then had to be withdrawn to below -- is that below the clavicle and clear of the shoulder joint?

BOHIN: That is what she's done.

MYERS: Yes, and the reason was because it was placed, it seems, in the wrong place?

BOHIN: Yes.

MYERS: Okay. And this is standard, is it, because not everyone is obviously familiar with the use of long lines? It's standard, is it, for five goes like that over this time and then withdrawing like this? That is standard?

BOHIN: Sorry, I missed that.

MYERS: Is that standard, five goes like this and then withdrawing in that way?

BOHIN: We don't know where it is going to end up so you have to withdraw it if it's not in the right place.

MYERS: In practice, is it standard, things like this, is this what happens regularly?

BOHIN: Yes.

MYERS: Five efforts, doesn't go in the right place, withdraw it?

BOHIN: Yes.

MYERS: And standard without fluid over that period of time?

That wouldn't be standard?

BOHIN: No, that's not standard.

MYERS: All right.

If we just go, please, to slide 31, so we're going to 9 June now, at 3.28 in the morning, slide 31, it's the nursing notes at 1355. Thank you.

(Pause)

So 3.28. So we're going into the morning here:

"Care taken over at 20.00 hours. All emergency equipment checked with respiratory support and fluids.

Nursed in air on CPAP."

Various descriptions but dextrose infusion at that time:

"Left hand and arm looked puffy at 21.00 and infusion stopped."

Then there's:

"Delay in the review and X-ray as sibling unwell and died. All doctors attending resuscitation. [Baby B] appeared jittery at this point. [And] BM and gas obtained by 5.1."

Do you see that, [Baby B] appeared jittery?

BOHIN: Yes.

MYERS: Later on bloods take place, rechecked bloods at 05.00:

"Jitteriness has settled. [Baby B] is very active."

That's what we have there. There's a reference to jitteriness after this has taken place; do you see that, Dr Bohin?

BOHIN: Yes.

MYERS: Did it appear to you that in the course of treatments with [Baby B], and it may be with others, there were problems sometimes getting the medical treatment or -- treatment they required at the right time because the unit was overstretched?

BOHIN: I think the unit had standard staffing for a unit of that size at that time. Nothing can ever be done immediately. That's impossible when you have -- whether in an adult ward or in a neonatal unit. You have lots of patients and there are demands

on you and you have to prioritise them. But I would say that the medical and nursing staff was standard for a unit of that size at that time.

MYERS: But standard, or however you would describe it, does it appear that at times it was overstretched?

BOHIN: No.

MYERS: In general?

BOHIN: No.

MYERS: You didn't see that?

Can we just go to clinical n1259, again in relation to [Baby B]. This isn't, I think, on the iPads. This relates to 15 June. So this is the period after collapse.

Just look at the upper entry, please. Just -- this is one from Dr Davis. 1.45 on 15 June. Written in retrospect:

"Cannula tissued. Due to workload on unit unable to site cannula immediately."

I'm not going to go through all of that, but in fact we carry on down, and we will:

"Tolerating feeds. Decision taken to increase feeds. BMs stable."

BMs, could you help us with that?

BOHIN: That would be blood sugar. That's the name of the machine. So that's the blood sugar.

MYERS: "CRP 3. Max CRP below 1 today."

CRP is a measure of inflammation, isn't it?

BOHIN: Yes.

MYERS: Which can itself be a marker of infection? It may be, take in conjunction --

BOHIN: It can. It's very non-specific, but it is a marker of inflammation.

MYERS: "Blood cultures negative"?

BOHIN: Yes.

MYERS: "Treated with", and it links to the antibiotics intravenously?

BOHIN: Seven days.

MYERS: Reference to:

"Dr Gibbs unable to site cannula. Can give..."

Is that --

BOHIN: Intramuscular.

MR JUSTICE GOSS: I think it may be:

"If unable to site cannula..."

BOHIN: "If unable to site cannula..."

MR MYERS: "If unable" --

BOHIN: "... give intramuscular cefotaxime and stop teicoplanin".

MYERS: All right. And we finish:

"Due to workload on unit unable to re-site cannula."

Yes?

BOHIN: Yes.

MYERS: I was just asking, whether from what you have seen in the case, and we may see this elsewhere, does it not indicate that the unit is overstretched as a result of workload?

BOHIN: I think -- no, I don't agree with that. Over time, because medical and nursing staff have been criticised for not doing things in a timely way, they have taken to writing -- and it's more common with nursing staff than medical staff -- "Due to workload, I haven't been able to do X, Y and Z" by way of explanation. But as I have said, doctors working on the unit have to prioritise their workload. The consultant had given an alternative way of administering these antibiotics and at this point it was only being used for antibiotic, it wasn't being used for anything else. So it was perfectly acceptable to give the cefotaxime by an intramuscular route.

MYERS: I'm not asking about the treatment. I was asking about workload. You have just said -- what did you say, that the nurses say what about their workload?

BOHIN: I said it's more common with nurses than doctors to write "Unable to do a particular thing because of the workload of the unit". That happens all over, not in the Countess of Chester. That's become a widespread thing.

MYERS: Right. And when they write that are you saying they are accurate when they say that or are they're just saying it?

BOHIN: I think partly it's a way of -- so that they are beyond criticism, really, so that if anyone questioned why this didn't happen in a timely way, you can explain that you were doing something else at the time.

MYERS: Putting aside what you're saying about they may just put it in there for whatever reason, why would someone question them why haven't they done this in a timely way? Why would that happen?

BOHIN: Well, depending on the number of babies you have, you are looking after -- you know, in a special care unit, you might be looking after four babies. You can't do everything to all of them at the same time. So something, at some point, will be delayed, and it's a way of saying: this was delayed because I was doing something else.

MYERS: Do you agree that in a unit is overstretched, there may be an increased risk of mistakes?

BOHIN: Yes.

MYERS: I want to move on with [Baby B], please.

I'm not going to go through the full history of breathing because I asked Dr Evans about that yesterday, and we've heard that. I'm not going to repeat all of that, but I'm just going to summarise. I can go to the detail if necessary.

There were breathing problems at birth with [Baby B], weren't there?

BOHIN: Yes, there were.

MYERS: And we looked yesterday at later on, on 19 and 20 June, where there was some desaturations recorded in the nursing notes; remember seeing that?

BOHIN: Yes.

MYERS: And a return to hospital on 14 July several days after discharge with other problems and a reference to mottling on that occasion?

BOHIN: Where she'd choked on a feed, yes.

MYERS: Yes. And the mottling was referenced then.

It is unsurprising that if there has been problems with breathing at the beginning and problems with breathing later on, that there may be some issues with breathing in between the two; do you agree?

BOHIN: No.

MYERS: Do you agree that a premature baby like [Baby B] can deteriorate without there being deliberate harm?

BOHIN: Yes.

MYERS: And she could deteriorate, like we see in this case at 00.30 on 10 June, without it being deliberate harm.

That's a possibility, isn't it?

BOHIN: It is a possibility.

MYERS: The fact there is a sudden deterioration does not of itself convert that into an air embolus, does it?

BOHIN: No.

MYERS: Air embolus is usually fatal, isn't it, from what we've seen?

BOHIN: It has a very high mortality.

MYERS: [Baby B] made a swift and good recovery, didn't she?

BOHIN: Yes, she did.

MYERS: So in terms of looking at what is more or less consistent with air embolus, a rapid and good recovery is less consistent with air embolus, isn't it?

BOHIN: Not necessarily, no.

MYERS: If it take very little -- I think you said a teaspoon yesterday.

BOHIN: No, I didn't say a teaspoon.

MYERS: That was suggested to you?

BOHIN: Mr Johnson said a teaspoon.

MYERS: That's right. You said maybe up to 5ml?

BOHIN: No, I said the reports in the literature which were extrapolated from adult and animal models suggested 3 to 5ml per kilogram of body weight.

MYERS: For adults?

BOHIN: Yes, and I noted some caution that that could not necessarily be extrapolated to neonates.

MYERS: But with neonates it is going to be a tiny quantity potentially, isn't it?

BOHIN: Whatever it's going to be, it's going to be a small quantity because the body weight was -- body weight was only 1.6 kilograms.

MYERS: So if you're saying 3 to 5 with an adult, it's going to be correspondingly overall with a neonate, isn't it?

BOHIN: Well, 3 to 5ml of an average adult weighing 70kg is going to be several hundred millilitres of air. These babies were 1.6kg, so 3 to 5ml times 1.6 is still a very small amount.

MYERS: And it's right to ask you to explain. If it's consistent with air embolus, could you explain why that is in [Baby B]'s case?

BOHIN: Well, what is consistent, the collapse?

MYERS: In terms of the recovery.

BOHIN: Again, there is very little on neonatal cases in the literature, but from experimental work with animals and in adults, it seems that it's the volume and speed of delivery that causes the clinical manifestations.

So in [Baby B]'s case, if the volume of air -- because I think her collapse was caused by air embolus -- if the volume of air was small or was infused rather slowly, then she would recover quite quickly because the effect of the air embolus would not be as devastating if a smaller volume was infused than if this a larger volume was infused.

MYERS: That's your explanation for how this could be an air embolus, isn't it?

BOHIN: Yes.

MYERS: I think you agree, if we look at the published studies and the literature, it's almost always fatal, isn't it?

BOHIN: Not always, no.

MYERS: It's almost always fatal?

BOHIN: No, most of the literature comes from adults, and although there is a high mortality, it isn't always because there are ways of treating it in adults.

MYERS: So are you saying that the fact that she survived and made a good recovery is absolutely neutral in terms of what it tells us about whether or not there was an air embolus; is that what you're saying?

BOHIN: No.

MYERS: It goes against it, doesn't it?

BOHIN: No.

MYERS: Are you deliberately excluding something which reduces the likelihood of air embolus, Dr Bohin?

BOHIN: No.

MYERS: Discolouration. Do you agree -- we've been there before with [Baby A], but let me ask you in the situation with [Baby B]. The fact of discolouration of itself doesn't go to establish air embolus, does it?

BOHIN: No.

MYERS: Can we look, please, at the clinical notes -- we looked at these yesterday -- at slide 233, which is 1282.

Thank you.

Can we just enlarge the central section where we get to the point where [Dr B] gave her description of what she could see, please, Mr Murphy. Can we see the words are:

"10ml per kilogram bolus given."

Just after "adrenaline not required", do you see that, Dr Bohin?

"Upon my arrival purple blotching, right mid-abdomen and right hand."

There's a full stop:

"Pink and active [full stop] in 40% oxygen. Sats 100%. H ratio (sic) 143."

Do you see that?

BOHIN: Yes.

MYERS: Let's take "purple blotchiness right mid-abdomen and right hand" first of all.

From that alone, do you agree, you certainly couldn't identify that as due to an air embolus if we're comparing it with the Lee and Tanswell description, could you?

BOHIN: No.

MYERS: No. In neonates when people talk about pink and active, as a rule, what are they talking about?

BOHIN: A baby who is well, who is moving about, and whose colour is uniformly pink.

MYERS: Yes. It's absolutely standard, the words "pink and active" describe a baby and its presentation, isn't it?

BOHIN: Yes.

MYERS: "Pink and active" does not describe a rash, does it?

BOHIN: No.

MYERS: Or skin discolouration?

BOHIN: No.

MYERS: It's the baby.

Do people in neonatal medicine commonly describe skin -- a rash as pink and active?

BOHIN: No.

MYERS: Now, yesterday Dr Evans, just towards the end of his evidence, was taken to this and he had the words pointed out to him:

"Upon my arrival purple blotching, R mid-abdomen and right hand, pink and active."

Do you remember that part of the evidence?

BOHIN: Yes.

MYERS: I'll just remind you of how it went. The question was:

"But in the next line it says right mid-abdomen and right hand pink and active; do you see that? [Dr B] interpreted her own handwriting for us this morning.

That pink and active wasn't read to you."

That was aimed -- I hadn't read it out:

"Do you see that?"

And Dr Evans went, "Yes, yes." And then he was asked:

"Is that consistent or inconsistent with the Lee and Tanswell?"

Do you remember him being invited to say whether it was consistent or inconsistent with Lee and Tanswell?

BOHIN: Yes.

MYERS: In effect inviting him to comment as to whether it's consistent with the colour of the rash. And Dr Evans' response was:

"It's a good point actually."

Yes? Do you remember him saying that?

BOHIN: Yes.

MYERS: "A good point."

Just to be clear, did you listen to the evidence of [Dr B] yesterday morning?

BOHIN: Yes.

MYERS: Let me remind you what [Dr B] said then, because she was asked about this. When she was asked about this section, she is the person who made this note, isn't she?

BOHIN: Yes.

MYERS: And she said:

"What I document there is that I have seen that there is purple blotchiness of the right mid-abdomen, the right hand, and the baby is pink and active."

Yes?

BOHIN: Yes.

MYERS: That's what she said.

Do you agree with Dr Evans' view that the term of the rash colour being pink and active is a good point?

BOHIN: I think what has been missed out is that there is a full stop there, and the two things are different. One is talking about purple blotchiness on the right mid-abdomen and right hand. There's an entirely different phrase to say the baby is pink and active.

I don't think they relate to the rash.

MYERS: Dr Evans was looking at the very same thing that you are, wasn't he?

BOHIN: Yes.

MYERS: Can I just ask this: do you agree when he says, in terms of describing the rash, that from these notes pink and active is a good point; do you agree with that?

BOHIN: No, I think that was a mistake.

MYERS: A mistake? Are you saying that would provide an excuse for Dr Evans?

BOHIN: I'm not excusing anything Dr Evans said.

MYERS: So you think he may genuinely have thought pink and active related to skin colour?

BOHIN: That is what was pointed out to him.

MYERS: All right --

MR JUSTICE GOSS: I think it's very difficult for one witness to comment on what another witness is thinking or why they are saying what they have said.

This witness's evidence, as I have understood it, is that [Dr B] has written:

"Upon my arrival purple blotchiness right mid-abdomen and right hand. Pink and active."

MR MYERS: Yes.

MR JUSTICE GOSS: Well, if it's pink and active, it's not --

"pink" describes the colour and "active" presumably describes the baby.

BOHIN: The baby moving about.

MR JUSTICE GOSS: The baby is active?

BOHIN: Yes.

MR MYERS: So in fact --

MR JUSTICE GOSS: It is what it is --

MR MYERS: I agree.

MR JUSTICE GOSS: -- and it's difficult for a witness to comment on it.

MR MYERS: Keeping it to you, Dr Bohin, to you it's patently obvious "pink and active" means the baby, isn't it?

BOHIN: "Pink and active" refers to the baby.

MYERS: And that's patently obvious to you?

BOHIN: It's obvious to me.

MR MYERS: Thank you, Dr Bohin.

MR JUSTICE GOSS: Thank you. Mr Johnson, I do not know how long you're likely to be but we have not had a break yet.

MR JOHNSON: I'm happy to have a break.

MR JUSTICE GOSS: I think we will have a break. Otherwise it gets rather late. We will just have a ten-minute break then. Thank you very much.

(In the absence of the jury)

MR JUSTICE GOSS: Thank you.

Right, thank you, doctor. If you can be back in 10 minutes and ready to continue. Thank you very much.

(12.09 pm)

(A short break)

(12.20 pm)

(In the presence of the jury)

MR JUSTICE GOSS: Yes, Mr Johnson.

MR JOHNSON: Thank you, my Lord.

Re-examination by MR JOHNSON

MR JOHNSON: Now, Dr Bohin, I want to clarify a few things from what you've said.

You were asked towards the beginning of the questioning whether you had started your exercise of writing your reports from the perspective of, in effect, accepting the theory that there has been deliberate harm. How do you view your role as an expert in a case?

BOHIN: I need to be completely independent and base my findings on the evidence presented to me.

JOHNSON: Would -- adopting that as a process that was suggested to you, would that be consistent with your view of your role in the case?

BOHIN: Yes.

JOHNSON: Would starting with a presupposed theory be consistent with your role?

BOHIN: I didn't start with a predispose (sic) theory at all.

I was asked to find out why these babies collapsed and that is what I did using the evidence that was presented to me with the clinical notes.

JOHNSON: Thank you.

MR JUSTICE GOSS: Now, I understand why it's happening, but it is happening. You're speaking more softly because the questioner is closer to you.

BOHIN: I'm sorry, yes.

MR JUSTICE GOSS: It's human instinct.

BOHIN: Apologies.

MR JUSTICE GOSS: Please keep your voice up. No, don't apologise. I do it sometimes.

MR JOHNSON: Now, you were then asked a series of questions about whether or not being in everyday clinical practice was a necessary prerequisite to being able to provide compelling expert evidence.

BOHIN: I was.

JOHNSON: And you said something that I think was slightly lost in response when you referred to consultant positions at tertiary centres. Could you just -- because I think I missed it. What did you say, please?

BOHIN: In a tertiary centre there is a very large team of consultants with a large raft of junior doctors under them, supporting them, and therefore the day-to-day clinical care is overseen by consultants, but things like the practice procedures are carried out by junior staff, and so I was just alluding to the fact that in smaller units the consultants are around and do more hands-on interventions than they would in a tertiary unit where there are lots of other competing requirements for consultant staff.

JOHNSON: And yet those doctors who aren't getting so much hands-on experience are, in effect, in charge of the more specialist units?

BOHIN: Yes, because in their time as juniors, they would have been in the junior doctor role carrying out the procedures. But by the time -- if you are a consultant neonatologist and you work in a tertiary unit, then your hands-on experience is, by definition, diluted because there are so many more of you. So you do less in terms of interventions and practical procedures, but you do oversee others.

JOHNSON: Thank you.

Can we go to S6 from the sequence, and this is -- relates, I think, to [Baby A].

This is the blood gas record. The questions that you were being asked about the PO2.

BOHIN: Mm-hm, yes.

JOHNSON: You told the jury this wasn't something that was reliable unless it was an arterial sample; is that right?

BOHIN: That's correct.

JOHNSON: So because this is a heel prick, a capillary sample, the result is not reliable, and that, you told us, is why under the AVC column, which is the third in from the left, one of those alternatives is always recorded?

BOHIN: Yes.

JOHNSON: If a C sample, a capillary sample, is not reliable, why is it recorded at all?

BOHIN: Because the other parameters are reliable. It's just the PO2 part of the blood gas that is not reliable. The other parameters are reliable.

JOHNSON: Is this just the standard list of results that come -- is spewed out by the machine once the sample has been submitted for analysis?

BOHIN: Yes, sometimes you can get other results as well that don't appear on this chart in terms of the salts in its blood as well, depending on the machine that the particular unit has. But these are the absolutely standard things that you would get in every blood gas.

JOHNSON: All right. So because you get a result, the result is recorded?

BOHIN: The result is recorded, but if it is a capillary or a venous sample PO2 -- although somebody writes it down, the PO2 is completely disregarded, it's not reliable.

JOHNSON: Thank you.

Now, you were asked about multiple attempts to put a line into [Baby A].

BOHIN: Yes.

JOHNSON: And you were also asked about something called negative venous pressure.

BOHIN: Yes.

JOHNSON: That, it was being suggested to you, could cause air to migrate through the line, is that right, into the circulation?

BOHIN: Only if the -- that could only happen if the end of the line was open to the atmosphere.

JOHNSON: Right. So let's deal with that, first of all. The evidence in this case from the treating doctors on that issue, what do you understand it to be?

BOHIN: When a long line or a central line was put in, that the doctors would have what we call primed the line. That means before the line is inserted, a fluid is put into that line, usually saline, so that there is no air left in that line.

I have one -- a line here actually if that would make it any easier.

So this -- I don't know if anyone can see it because it's so tiny. This is an umbilical venous catheter.

You can see it's really small. The long lines are actually smaller and have a little wire in but to all intents and purposes it's the same, and it has a little plug on the end which a syringe can be attached to.

So you attach a syringe of saline to this, and inject saline until you get saline out of the end. So that is then full of fluid -- it's not full of air, it's full of fluid -- and then you insert it into the baby's belly button. Once it's in, you suction on a syringe to get blood back so that you know you're in a blood vessel.

So that tube is never left without anything in it.

It's never left with air in it. You fill it with air -- fill it with saline before you start the procedure.

JOHNSON: Yes.

BOHIN: And there is something on the end of this, be it a bung, like this little cap, or a syringe. So it's never put in and left open to the air so that air can be inadvertently sucked in.

MR JUSTICE GOSS: Sorry, is that -- that's an umbilical venous catheter?

BOHIN: Umbilical --

MR JUSTICE GOSS: You did earlier just refer to long line.

Doctors would have --

BOHIN: A long line -- it looks pretty similar except it's got a wire in it that you then remove. But the actual process is -- of inserting it is exactly the same. You still prime the line before you put it in, so there is no air left in that tiny plastic tube.

MR JUSTICE GOSS: All right.

Could you all see that? I know you're some distance away.

BOHIN: It could be handed to them.

MR JUSTICE GOSS: Do you want it to become an exhibit?

MR JOHNSON: Perhaps it could be exhibited and then if reference needs to be made to it at a later stage --

MR JUSTICE GOSS: Well, I don't think -- the jury don't want to look at it now, but if it could be formally exhibited then it's in evidence so that if at any time -- once something becomes an exhibit, it's an exhibit until the very end of the case. So it's there if anyone wants to look at it.

MR JOHNSON: Just dealing with that then, is -- is the line itself see-through or not? Can you -- if you're looking -- just looking at it, if it was going into a child's belly button, would you actually be able to see through the line that is on the outside?

BOHIN: No. No.

JOHNSON: Next you were referred to a paper, a medical paper written by Gordy and others.

BOHIN: Yes.

JOHNSON: And a passage was put on the screen from J52039. This referred to air embolisms; do you remember that?

BOHIN: Yes.

JOHNSON: Now, were you about to give us an explanation that was then interrupted by a further question in relation to something you were going to say about this.

BOHIN: Sorry, I can't recall.

JOHNSON: You can't remember, all right.

Well, let me just see because it may be that I got completely the wrong impression of what you were saying and it wouldn't be the first time that I've got the wrong impression about something.

Part of this paper was read to you where it says:

"Air embolism is a rare but fatal occurrence and may result from a variety of procedures and clinical scenarios."

So this is taken at the top.

BOHIN: Yes.

JOHNSON: "It can occur in either the venous or arterial system depending on where the air enters the systemic circulation. Venous air embolism occurs when gas enters a venous structure and travels through the right heart to the pulmonary circulation."

That means the right side of the heart. So that's the venous return to the heart; is that right?

BOHIN: That's correct.

JOHNSON: And then goes out to the lungs, the pulmonary circulation; is that right?

BOHIN: That's correct.

JOHNSON: "Conditions for the entry of gas into the venous system are the access of veins during the presence of negative pressure in these vessels."

And should we read into that conditions for the "innocent" entry of gas into the venous system is the access of veins?

BOHIN: Yes.

JOHNSON: So if you were injecting air into the system, you wouldn't need negative pressure?

BOHIN: No.

JOHNSON: Is this a study on adults or a study on neonates/babies?

BOHIN: It's an adult study.

JOHNSON: Right. Is there a difference at all between the circumstances relating to neonates and the circumstances relating to adults?

BOHIN: Yes. Firstly, obviously, their size, and the reason for the intervention in the first place that required these adults to need a central line put in place.

So on occasion in this study adults were undergoing surgery, sometimes neurosurgery, and they needed to have a central line put in. Sometimes these adults were already very ill and were in an intensive care unit. In some of these cases, because the adults were having neurosurgery, they were sitting up. So they were in an unusual position to have a central line inserted.

None of those things can be extrapolated down to neonates who are always on their back or on their front.

They are always lying down. They are never sitting up because they can't sit up.

And obviously neonates are much smaller and the lines, as I have just shown you, are really very tiny, whereas in adults they are much bigger lines because adults are bigger and you need to be able to get an increased volume of fluid or anaesthetic or drug or whatever into the adult. So the lines themselves are much bigger.

JOHNSON: Have you ever heard of a neonate receiving an air embolism via the negative pressure theory, if I can call it that?

BOHIN: Never.

JOHNSON: Thank you. Can we move to J1353 next, please.

MR JUSTICE GOSS: Sorry, in relation to that last answer, can I just be clear. You have never heard of a neonate suffering an air embolism by reason of negative pressure?

BOHIN: That's correct.

MR JUSTICE GOSS: Is that both in clinical practice, in relation to research in literature?

BOHIN: I've not seen it in relation to the research in literature with neonates, no.

MR JUSTICE GOSS: Or any educational events that you have attended?

BOHIN: No. I think that's important because if things are common and worrying and serious, as I said before, an alert would be issued so that everybody dealing with neonates would be aware of potential problems.

So if this had been a common or -- well, even a rare but -- like a never event that shouldn't happen, people would be mindful of it because an alert would come out, and there has been no alerts. And I have not ever come across it in conferences or in any educational courses that I have attended.

MR JOHNSON: Can we go at 1353 -- thank you. You're ahead of me, as always, Mr Murphy.

This note relates to issues concerning [Baby B], I think. Let me just check that I've got that right and it's not [Baby A].

MR JUSTICE GOSS: It is. Female sex on the top left.

Sex: female.

MR JOHNSON: Okay.

So if we can go back to -- so this is [Baby B]?

BOHIN: Yes.

JOHNSON: This was pointed out to you in cross-examination where Melanie Taylor recorded at 18.58 on the 8th, so in other words, just to put that into context, about an hour and a half or so before [Baby A] collapsed and died?

BOHIN: Yes.

JOHNSON: And about 28 or 30 hours before [Baby B] collapsed, this records that [Baby B]'s UVC was in the wrong place, removed. The SHO was informed:

"Tried to cannulate [Baby B] because the registrar was busy. Failed attempts. No fluids running for a couple of hours. Doctors aware."

So here we have [Baby B], at about the same time that [Baby A] wasn't getting fluids, also not having fluids?

BOHIN: Yes.

JOHNSON: Okay. First of all, did [Baby B] collapse within the next hour or so after that?

BOHIN: No.

JOHNSON: Secondly, is there any link between this event and her collapse 28 hours or so, 30 hours later?

BOHIN: No.

JOHNSON: Thank you. I want to go next, please, to J1339.

Now, to put this into context, these notes concern the attendance of [Baby B] at the Countess of Chester on 14 July 2015. Okay?

BOHIN: Yes.

JOHNSON: So in general terms, about 5 weeks after she'd been born --

BOHIN: Yes.

JOHNSON: -- give or take.

You were asked in passing about this when it was suggested [Baby B] had stopped breathing, and you, I think, in passing in your answer referred to her having choked on a feed. I just want to deal with what actually happened, what was being reported by mum as actually having happened. Okay?

So we can see that this is a note of Dr Gail Beech, whose name we know. We know that [Baby B] had attended at A&E with her mum about an hour and a half before this, I think. Four hours, I'm sorry. Four hours. So between attending at A&E and actually seeing the doctor was 4 hours. And we can see Dr Beech's note, which recounts a history taken either from mum or from grandma.

BOHIN: Yes.

JOHNSON: Seen with mum and grandmother.

Could you just interpret the note for us, please?

BOHIN: Yes:

"History noted with thanks. Ex pre-term infant.

Discharged from the neonatal last week. Yesterday evening had lactulose [which is something given for constipation, a medication,] and immediately afterwards a feed.

"During the feed had a 'mucousy' [in inverted commas] vomit..."

"Mucousy" means just kind of vomit mixed with kind of spit and saliva:

"... which mum now thinks is the lactulose dose."

Lactulose is a clear medicine and it's very sugary and sweet and it's quite thick and gelatinous, so actually it could look mucousy:

"Mum thinks it's the lactulose dose.

"Does not normally feed so soon after lactulose dose.

"Struggled to catch breath for a few seconds afterwards but no colour change, then fed again. Has taken a feed since with a couple of small vomits but mum feels it was because of position [of] milk came out too quickly."

JOHNSON: "And milk".

BOHIN: "... and milk came out too quickly... by Dr Prior who agrees."

JOHNSON: "Witnessed by Dr Prior".

BOHIN: "... witnessed by Dr Prior who agrees. Otherwise completely well yesterday. No temperature/rashes/cough/diarrhoea. Has been sneezing recently. Waking for feeds and taking required volumes.

Passing urine [tick], bowels open [tick]."

JOHNSON: Is that note, that history, suggestive of this child having breathing problems?

BOHIN: No, in fact it's quite common for newborn babies, whether they're term or pre-term, to have quite a lot of difficulty with co-ordinating breathing and swallowing.

And so if you're taking a feed and you need to swallow, quite often babies will get the two confused.

It is more of a problem in premature babies, but usually they have a cough and clear whatever is there and then swallow it.

JOHNSON: Okay. Towards the end of the questioning the issue of discolouration was investigated with you.

BOHIN: Yes.

JOHNSON: And you were asked in the context of [Baby B] whether the fact of discolouration -- what was suggested to you was that the fact of discolouration in itself doesn't go to establish air embolus, does it? You replied no, and the question I'm going to ask you is why not, and why are you suggesting that [Baby B]'s case is a case of air embolus?

BOHIN: There are no single diagnostic factors of or signs of air embolus. What we know from the limited studies that have been carried out in neonates is that air embolus is associated not only with this very florid description of the colour changes seen, but also of a collapse with low heart rate, desaturation and a stopping of breathing.

So the colour change on its own is not diagnostic of air embolus, but I was -- in making my diagnosis I put the colour change described by the clinicians into context with the other things that were happening with [Baby B] at the time, ie her sudden unexpected profound collapse.

MR JOHNSON: Thank you very much.

Does your Lordship have any questions?

MR JUSTICE GOSS: No, I don't. Thank you very much indeed.

Thank you, doctor. That completes your evidence at this stage, but as you know, you will be coming back to give evidence later during the course of the trial.

Thank you.

So if you could just leave the -- I don't know whether -- it's already been produced, has it?

BOHIN: Yes.

MR JUSTICE GOSS: It's still in there, is it?

BOHIN: Yes.

MR JUSTICE GOSS: Could you leave that with us, please, and we will give it an exhibit number.

BOHIN: Yes, of course.

MR JUSTICE GOSS: Thank you.

MR JOHNSON: My Lord, if I may, I'm going to recall Claire Hocknell, the analyst, formally to deal with the corrections to this sequence of events.

MR JUSTICE GOSS: Thank you. Yes, thank you.

MR JOHNSON: I have thought about whether this needs to be done formally. I think it's probably better that there is a record of exactly what we're doing, if that's --

MR JUSTICE GOSS: It won't take that long.

MR JOHNSON: It will take about 5 minutes.

MR JUSTICE GOSS: Exactly. All right. Thank you anyway.

MR JOHNSON: I know it could be done by agreement, but it's better that everyone knows where we are.

MR JUSTICE GOSS: And it makes sure everyone is aware of what the corrections are and they can -- all right.

So I don't suppose you've ever not had the experience of some long document that's been prepared that you've had some part in preparing is found to have some clerical errors in it, and I think this is what we are going to do now. Some clerical errors have been picked up in the compilation of the sequence documents and they are going to be corrected now.

MS CLAIRE HOCKNELL (recalled)

Further examination-in-chief by MR JOHNSON

MR JOHNSON: Mrs Hocknell, I think you have been consulted -- I think you saw a number of errors yourself as we went through the process at the beginning --

HOCKNELL: That's correct, yes.

JOHNSON: -- of the [Surname of Babies A & B] children's cases, and you have had one or two other issues raised with you, primarily concerning handwriting of doctors?

HOCKNELL: That's correct, yes.

JOHNSON: Given that you're not a pharmacist, you weren't able read to necessarily every single word. So the one or two that have been corrected by the witnesses themselves have now been incorporated into your work --

HOCKNELL: That's correct.

JOHNSON: -- which is embodied in this sequence; is that right?

HOCKNELL: That's correct.

JOHNSON: Thank you. So starting with 183, please, can we get -- it's your summary that has been corrected.

HOCKNELL: Yes.

JOHNSON: The word "finishing" now appears where formerly was the word "starting"?

HOCKNELL: That's correct, yes.

JOHNSON: Thank you. At 185 next, please.

MR JUSTICE GOSS: These corrections have been added --

MR JOHNSON: It's been done, yes.

MR JUSTICE GOSS: So you needn't worry that you have got to change them. They have been changed. They're just being drawn to your attention as to what -- you will remember the starting and finishing one because we had that in evidence, and it was obviously an error because when you looked at the original it was completely the wrong one. Not critical, but...

MR JOHNSON: Now, this one -- yes. It's -- where the cursor is, under the word "on", I think you had previously typed "oi".

HOCKNELL: Yes, I think it was a typo on my part.

JOHNSON: Yes. So that's been corrected. Thank you.

And then so far as -- there are three other corrections. The next is 246 which was a text. It formerly appeared as a text from [Nurse A] to Lucy Letby and that was the wrong way round. So that has been reversed.

HOCKNELL: That's correct, yes.

JOHNSON: Thank you. Slide 284 next, please. Formerly the sender was Lucy Letby and the receiver Jennifer Jones-Key.

That has been reversed.

HOCKNELL: That's correct, yes.

JOHNSON: And then finally slide 354, please. Formerly the sender was Lucy Letby and receiver [Nurse A], and that too has now been reversed.

HOCKNELL: That's correct, yes.

MR JOHNSON: Thank you. Thank you very much.

I don't know if there are any questions?

MR MYERS: No questions, thank you.

MR JUSTICE GOSS: Thank you very much.

(The witness withdrew)

MR JOHNSON: My Lord, there is, I think, one more issue to deal with so far as the case of the [Surname of Babies A & B]s is concerned. Just to make sure I don't use the wrong document, I wonder whether we could start -- end slightly early and start again slightly early, just to eliminate the possibility of a mistake?

MR JUSTICE GOSS: Yes, certainly. I have committed to another work engagement during lunch, but that's fine.

It starts at 1.15 and it will finish by 2 o'clock. So if we could say 2 o'clock then or 2.05.

MR JOHNSON: Yes, perfect, thank you.

MR JUSTICE GOSS: Thank you.

Nothing to do with this case but I've got to work through lunch.

So could you be ready, please, to resume at 2.05, ladies and gentlemen.

And remember that today -- some of you may have remembered -- we're finishing early, slightly earlier than normal. No later than, I'll say, 3.40.

(In the absence of the jury)

MR JUSTICE GOSS: Yes, thank you very much.

(12.54 pm)

(The short adjournment)

(2.08 pm)

(In the absence of the jury)

MR JOHNSON: My Lord, before the jury comes in, if I can explain what's going to happen next.

By agreement between the parties we are giving the jury a summary of what Lucy Letby said in interview about these two cases.

This is not a substitution for the interviews --

MR JUSTICE GOSS: No.

MR JOHNSON: -- which will come at the end of the case in the traditional way. One of the reasons for that is because what is said on various cases has to be seen in the context of the development of the interview.

I think we all agree that it's helpful for the jury to at least have a summary of what she said at the time at this stage. They are not going to be given the summaries in writing. I'm simply going to read them.

But it will illuminate some of the issues that have been dealt with.

MR JUSTICE GOSS: And the script that you're going to read is obviously agreed?

MR JOHNSON: It is.

MR JUSTICE GOSS: Thank you very much indeed. I think that's very helpful. Then at least the jury will have it in mind because we are compartmentalising these cases.

Thank you. That seems very sensible.

So if you will -- go ahead, Mr Myers.

MR MYERS: I didn't mean any discourtesy but in fact I was about to have a word with Mr Johnson along the lines your Lordship is about to speak to him, I think.

MR JUSTICE GOSS: Yes. Just to explain what is happening with the caveats in relation to it. So they aren't to take this as an exhaustive account?

MR JOHNSON: No.

MR JUSTICE GOSS: It is merely a summary to highlight for them, basically, what was being said at the stage of interview?

MR JOHNSON: Thank you.

MR JUSTICE GOSS: Thank you.

Yes, please.

(In the presence of the jury)

MR JUSTICE GOSS: Yes, I'm sorry we are a few minutes late.

Mr Johnson?

MR JOHNSON: Thank you, my Lord.

Ladies and gentlemen, when I introduced the case to you a couple of weeks ago now, you will remember that I gave you an outline of each of the cases and I also told you, really probably in about 2 minutes, what Lucy Letby said in interview about each of the cases.

But what we're going to do now is, by agreement between the prosecution and the defence, I'm going to read to you an agreed summary of the interviews, but only so far as they are related to [Baby A] and [Baby B], okay.

At the end of the evidence, when we've dealt with all the cases, you'll get the interviews in one go, but those will cover everything, and we have agreed between us that it may help you at this stage, while you have still got the evidence relating to [Baby A] and [Baby B] clear in your minds, it may help you to know what Lucy Letby said about it.

So this is only a summary. We won't give you this in writing because you will get the full interviews in writing in due course. Okay?

MR JUSTICE GOSS: So it's a matter for you, as always, to decide whether you want to make any notes of this, but bear in mind this is nothing more than a summary.

You'll get much more detail later on in writing in front of you. So it's up to you. Some of you, I know, take a lot of notes as you're going along. My only word of caution about that is that sometimes you will, by the end of the case, have literally a book full of notes or several books of notes and it's sometimes difficult to go back and work through them. That's why I suggested maybe just what you consider may be important features as to what is to note. So you may just want to note bullet points. You may just want to listen, absorb it, and you will remember it. As I say, you will get them in detail, in writing, later on.

Thank you, Mr Johnson.

Summary of interview with Lucy Letby re [Baby A]

(read)

MR JOHNSON: Thank you, my Lord.

So dealing with the case of [Baby A] first, the first interview took place on 4 July 2018. And Lucy Letby was given a chance to go through the relevant notes of [Baby A]'s treatment. And she was asked, and this is a direct quotation:

"Do you specifically remember this shift with [Baby A]?"

And she replied:

"Yes."

She said that the handover was at about 8 pm at his bedside in nursery 1. She said that the nurse caring for him, Melanie Taylor, was sterile and in the process of connecting or drawing up some fluids to connect via a long line that [Baby A] had had inserted during the day, and she recalled that the long line had been set up by the doctor.

Lucy Letby said that whilst she was receiving a handover from Nurse Taylor, she checked the fluids and that Nurse Taylor had connected them. She was asked this question:

"So did you retrieve that bag for [Baby A]?"

And she replied:

"Not that I recall. I recall that Mel already had the bag out."

She said:

"Soon after we noticed that [Baby A] had gone pale and mottled in his skin."

She said:

"We called Dave Harkness over and initiated airway support as [Baby A] was apnoeic. A crash call was put out and Dr Jayaram attended."

She said that full resuscitation attempts followed.

When asked about the handover, Lucy Letby described [Baby A] as:

"A little bit jittery in appearance and in his limbs."

She thought this may have been due to low blood sugar and she said he was due to have his blood gas and blood sugar levels checked shortly after she, that is Lucy Letby, came on shift.

She then said:

"Mel was there with the fluids when I came to get handover and we made that a priority, to check the fluids, and got those set up and running."

The question was this:

"Okay. And again, why was that? Was that a priority?"

And she replied:

"Because he hadn't had his cannula, his peripheral cannula, inserted in the day. He had a UVC that they weren't happy to use and he had a long line inserted that the doctors had confirmed we could use. So I think he'd gone a few hours without any fluids."

And question:

"Okay, and is that normal? Is that a cause for concern?"

She replied:

"It's not ideal, but when you have a new baby and it's -- you have -- you can have problems obtaining access to be able to give fluids."

She then said that the doctors had confirmed that the long line was ready to use and fluids were run. She said that [Baby A]'s colour changed and he became more pale after, she said, Nurse Taylor had connected the fluids.

She then said:

"It was fairly soon, within minutes, I think."

She was asked:

"Thirty minutes? Two minutes?"

And replied:

"I don't know, maybe 5 minutes."

She remembered that she was at [Baby A]'s bedside checking equipment as it happened. She said that it was usual to receive

handover, and then check all emergency equipment at the cot side.

Lucy Letby said that Melanie Taylor remained in the room at this point. She couldn't remember what Nurse Taylor was doing, but thought that she may have been writing her notes. She said that Dr Harkness was in the room dealing with [Baby B] and said that Caroline Bennion was also in the room.

She said the mottling she saw was:

"... blotchy, red and purple, almost a rash-like appearance."

Which Lucy Letby thought could be a sign of:

"... infection, low blood sugar, being cold or low blood gas."

It appeared on his hands and feet and it was to the left side of his body where his line was, but he was centrally pale.

She was then asked this question:

"Was it bright red? Was it dark red?"

And she replied:

"I think he was more pale than the mottling, than the little areas of the mottling, but predominantly it was paleness."

And the question:

"Did anyone else see that mottling on [Baby A]?"

Her answer:

"Yes. I think it was there when I called Dave, and Caroline and the other nursing staff came in."

Then the question:

"And did they say anything about it to you?"

Answer:

"Yes. We were advised to stop the fluids on the long line straight away."

Lucy Letby then said that when she went to [Baby A], she checked that the CPAP mask was still in place. She checked to see if the machine was running and if he was breathing. The machine was

still running and she found that he was apnoeic and not breathing.

Lucy Letby then said:

"And then, straight away, Dave was in the room.

Mel Taylor and I called him over from what I recall."

She said that Mel Taylor and Caroline Bennion were still in the room and that:

"We were advised to stop the fluids on the long line straight away."

She couldn't -- Lucy Letby said that she couldn't recall [Baby A]'s resuscitation, but that at some stage Dr Jayaram attended the resuscitation which Dr Harkness had commenced. She said that [Baby A] then passed away.

Lucy Letby said it was awful and she agreed with the officer that it had been difficult for her. In terms of coping mechanisms, Lucy Letby said the team supported one another with it. She was asked who else she would have spoken to and couldn't specifically recall, but felt that she would have told her parents and her friends.

She couldn't remember if she'd passed [Baby A] to his parents after his death. She believed that mum's health problems were mentioned at the debrief held by Dr Jayaram and, when asked in interview, she thought it may have been a problem with either the line or the fluid attached to the line.

She couldn't recall for definite who had attached it, but believed it had been Melanie Taylor as she was sterile.

In considering the notes, she agreed that she'd completed [Baby A]'s handprints and footprints, with the consent of parents, for his memory box and had also taken a lock of his hair and the handprints and footprints of his sibling as the parents had wished.

She said photographs were taken with the unit camera and the parents' phones, and she found the process quite a nice thing to do for the baby and the parents, as if it was a way of giving the parents memories.

She said there was a bereavement checklist to go through when a baby had died, so the nurse would tick things that had been done such as handprints, footprints and photographs.

She discussed the nursing rota and shifts. She confirmed that she could often be called on a morning and asked to work overtime that night. On a week-to-week basis she could receive calls stating the unit was short of staff.

She was asked further questions on 10 June 2019, moving on a year. She agreed that she had completed training on the administration of medications via long lines in May of 2015. She didn't think it contained anything specific on the risk of introducing air.

In November 2020, on 10 November 2020, she was interviewed again. She agreed that she'd been taught to prime lines so that they didn't have air in them and that it would be hard to push air into a line deliberately.

She denied having done so via either [Baby A]'s UVC or the long line, and she said that she didn't know exactly what an air embolus was. She had been taught to clear the line and make sure there wasn't any air in it as it could lead to an air embolism.

She had been trained how to avoid air embolism due to the danger they present and she was unaware of the physical effects of an air embolus and any changes to the appearance of the skin.

She was questioned about Facebook and confirmed that she had an account at the time, which she'd access via her phone or a tablet. She explained that her relationship with the parents on the unit was only ever professional, either face to face in the unit or by telephone when necessary. She couldn't explain why she'd searched for [Baby A]'s mum, [Mother of Babies A & B], on three separate occasions in June 2015. She couldn't remember conducting the searches or why she'd done it.

When asked about a further search in September 2015, she thought she may have been looking for an update on [Baby B] as the nurses thought about babies on the unit and talked about them.

So that's [Baby A].

Summary of interview with Lucy Letby re [Baby B]

(read)

MR JOHNSON: In relation to [Baby B], Lucy Letby was first interviewed about [Baby B] on 4 July 2018. She'd been given access to the notes of [Baby B]'s treatment.

She noted the administration of Babiven and lipids from 00.05 on 10 June 2015. She agreed she'd signed that entry, which ideally should have been co-signed.

She confirmed that [Nurse A] was [Baby B]'s designated nurse.

She said that she couldn't remember the shift with any clarity, but did remember [Baby B] displaying some mottling that looked a little bit similar to that seen on [Baby A] the day before. It was more extensive and covered more of her body in comparison rather than just the limbs like [Baby A]. [Baby B]'s was like a purply red patchy rash-like appearance.

Lucy Letby thought that [Nurse A] may have alerted her to [Baby B]'s appearance, and Lucy Letby said that she was either in the room or at the nurses' station.

[Baby B]'s appearance was observed before any resuscitation began. Lucy Letby said she couldn't recall if [Baby B] was attached to a monitor, but assumed she would have been. She didn't recall any alarm sounding. She said she didn't have any recollection of her interactions with [Baby B], nor did she recall having any concerns for [Baby B]. She couldn't remember in which room her designated baby was that night, but [Baby B] had been in nursery 1 and she would have attended to [Baby B] if [Baby B]'s designated nurse was on a break or engaged with another baby.

She confirmed that if a nurse caring for that baby is doing something else, then a member of staff may do the drugs for the baby.

She confirmed that she would have handled [Baby B] to an extent to give her medication and attach those lines.

She was asked about connecting via a long line or a UVC, and this question was asked:

"Does it need two people to do that?"

She answered:

"Yes."

And the question:

"Because of the protocol or because physically you need two people?"

She answered:

"So it's protocol that we would both go to the patient and connect it together."

She said she'd taken the blood gas readings while another nurse would have been operating the Neopuff machine, and then the following exchange took place:

"Question: At that time then do you remember how [Baby B] was, how she was presenting?"

She responded:

"I don't recall at that time, but looking at this, it was happening during Neopuff, so she was obviously being Neopuffed by somebody while I was taking the blood gas."

The officer then said, "Right", and she said:

"And the blood gases aren't great, so..."

And the officer said:

"Okay. So does that say to you that's post -- post-collapse?"

She replied:

"Or during, yes."

Lucy Letby said:

"The only concern primarily were the patients (sic)

and how they were coping with that. Obviously they'd already lost [Baby A], and then to have to see [Baby B] go through resuscitation."

She didn't recall how upset [Baby B]'s parents were and the following exchange took place. She said:

"Just I know that they had waited a long time for [Baby A] and [Baby B], that they were much-wanted babies."

The officer asked:

"Who told you that?"

She replied:

"Answer: It was known through the handover you have. So when you have a nursing handover you have a little bit of background as to the parents and the pregnancy and things."

"Question: Does that sort change the way you deal with them or affect the way you deal with them?"

"Answer: No. I would like to think you treat all parents the same, but obviously you just bear in mind what we have they've gone through to get to that point.

But I think the care you would give them and their babies would be the same as any other parent."

So, my Lord, that's the summary at this stage, and there will be a lot more information in due course.

MR JUSTICE GOSS: Yes, thank you very much.

MR JOHNSON: We are now going to move on to the case of [Baby C].

MR JUSTICE GOSS: Thank you.

(Pause)

MR JUSTICE GOSS: Yes.

MR DRIVER: My Lord, as my learned friend Mr Johnson just indicated, we are now going to move through the indictment to count 3 and the evidence in support of the allegation of murder in relation to [Baby C], who died on 14 June 2015. So we're moving from the events of the 8th to 11 June 2015 and to the days thereafter.

But as your Lordship knows, and perhaps the jury ought to be informed, although we've substantially heard the evidence in relation to [Baby A] and [Baby B], for good logistical reasons, some expert evidence has been put back to later in the case. So we shall return to them in due course through the evidence of Professor Kinsey, the paediatric haematologist, and the pathologist, Dr Shukla and Dr Marnerides. But with those exceptions we will now move on to the case of [Baby C].

Your Lordship should have a bundle of statements, about 26 pages in there. I'm grateful.

Witness statement of [Mother of Baby C] (read)

MR DRIVER: Members of the jury, the first evidence in relation to this count to be placed before you comes in the statements of [Baby C]'s parents, which I'll now read to you.

The first statement is by his mother, [Mother of Baby C], dated 13 January 2019. And it reads:

"I am [Mother of Baby C]. I provide this statement at the request of Cheshire Police. It gives a family history and

describes my experience of the neonatal unit at the Countess of Chester Hospital and provides details of the tragic death of my son, [Baby C].

"I became pregnant to my husband [Father of Baby C] in early December 2014. This was my first pregnancy. At 20 weeks I went for an anatomy scan at the Countess of Chester Hospital. During the scan, the sonographer informed us that although everything appeared normal, the baby looked very small. At this point one of the foetal medicine midwives, named Julia, came in to see out. She explained she had liaised with one of the consultants, Sara Bringham, with regards to my pregnancy and it had been recommended that I had an amniocentesis to check if there were any genetic problems with my baby."

And the statement goes on to explain the nature of that test, and it's a test to check if a baby has any genetic or chromosomal conditions and involves the removal and testing of a small sample of cells from the, amniotic fluid.

[Mother of Baby C]'s statement goes on:

"I attended an appointment for an amniocentesis with the consultant Jim McCormack. Jim became my consultant from this point on and saw me through the remainder of my pregnancy.

"I was around 21 weeks pregnant when I had the amniocentesis. The results came back about a week later and showed there were no genetic problems with my baby.

He was small but it was not known why.

"From this point on I underwent regular scans which were conducted by Jim McCormack. They were held approximately fortnightly. [Baby C] had growth scans and blood flow checks.

"Around 28 weeks into the pregnancy things took a bit of a turn for the worse. [Baby C]'s growth began to falter and there were signs that the end of my pregnancy was getting close. I had been working up to this point, so, as a result of these changes, we decided that it would be better if I stopped working as a precautionary measure.

"I was given steroids to help mature the baby's lungs, as it was expected that the delivery would take place quite soon. The steroid was administered by an injection and I was given a further dose of steroids around 24 hours later. I was now being monitored twice weekly.

"On Friday, 5 June 2015, it was found that my blood pressure had raised a little. I underwent a blood test to check my liver

function, and although it wasn't too bad, it had always been normal and it wasn't anymore.

Mr McCormack, the consultant, was on holiday. So Sara Brigham saw me and she admitted me to hospital that day as a safety precaution as she felt I would give birth in the next few days and she wanted me to be in a position where I could be closely monitored. I went home with my mum, packed some clothes and returned to the Countess as advised.

"That night, when being monitored, it was found that [Baby C]'s heart rate had dropped at around 2 in the morning. I was moved down to the labour ward because concerns had grown and it was felt that I may need to deliver my baby. On this occasion nothing happened. So in the morning I was moved back to the antenatal ward.

I continued to be to monitored over the weekend.

"On Monday, 8 June 2015 Mr McCormack visited me and told me he would have delivered my baby on 15 June had he been present, but now he felt, looking at the current monitoring, it would be best to carry on with close monitoring. He told me that if my condition returned to how things were on the Friday, then the baby should be delivered.

"The monitoring continued on the 8th and the 9th and then, on Wednesday 10 June, I went for a scan early in the morning. Mr McCormack performed my scan and told me there was no blood flow through the cord, and in fact it had started to reverse, which is a critical sign that you could be in danger of having a stillborn baby and the baby needed to be delivered that day.

"I phoned [Father of Baby C] to arrange for him to come to the hospital before I was taken to the labour ward.

I was on the ward for most of the day and was given various drugs which helped to support the baby. One was magnesium sulphate, which matures the baby's brain.

"Somewhere between 15.00 and 15.30 hours I was taken to theatre where Ms Rowe(?), an obstetrician, conducted a caesarean section procedure and delivered our baby, [Baby C].

"The birth was extremely stressful. There were several paediatric personnel within the theatre.

"[Baby C] was taken to the Resuscitaire on delivery in order to take care of him because he was so small and fragile. Sally [which is one of the doctors] came and told me she had intubated

him and placed him on a ventilator, but at that time they weren't necessarily concerned for him.

"I was treated and returned to the post-natal ward.

I wasn't able to visit [Baby C] for a number of hours as I needed to be able to stand without assistance. Being able to stand unaided was also hindered by the fact that I had been given a spinal anaesthetic as part of the caesarean.

"I estimate it to have been around 8 or 9 o'clock on 10 June when I went down to see [Baby C]. He was now in nursery 1. He was tiny. He only weighed 800 grams.

"By the time I went to see [Baby C], the hospital staff had already taken him off the ventilator. He was given alternative breathing support in the form of continuous positive airway pressure, CPAP."

It then continues to describe the events and says:

"For the first time on 11 June, which was a Thursday, [Father of Baby C] and I had small periods of time where we could hold [Baby C]. I would say the neonatal nurse who was on most of the time was Jo [that is Joanne, Williams]. She was very supportive and reassuring and was able to answer all of our questions.

"Friday 12 June was much the same. The only consultant we had any dealings with was John Gibbs.

"The following day, Saturday, 13 June, [Father of Baby C] and I spent all day on the neonatal unit. They had taken [Baby C] off the CPAP and put him on a very light flow of oxygen. [Baby C] did appear to be a lot calmer, having been a little agitated the day before. [Baby C] had a gastric tube fitted which went into his stomach.

When they drew fluid from his stomach, I recall it being a black colour. They were unable to explain why this was, and as far as I am aware, it wasn't queried with the Liverpool Women's Hospital.

"The staff remained positive with [Baby C]'s progress.

[Father of Baby C] and I also had a cautious positivity. We weren't being unrealistic, but we could see he was doing all that could be expected of him, and his oxygen levels were being weaned down because he was doing so well.

That really was the state of play for the 13th.

"I was last on the neonatal unit around 10.30 that night when I had taken some milk down. There was a nurse on the unit looking after him that I had not seen before. She had dark hair and I felt she was a little abrupt with me, and after this I didn't see her again.

"Initially, when [Baby C] was placed in nursery 1, there were other babies with him. Now, on Saturday the 13th, he was the only baby there.

"Having gone to bed, I was awoken at 23.20 hours by one of the midwives from the post-natal ward. She had been contacted by one of the neonatal nurses who were requesting that I went to see [Baby C] urgently. She wouldn't tell me why.

"I ran down to the neonatal unit where I could see CPR was being performed on [Baby C]. It was explained to me that his heart rate had suddenly dropped and he had stopped breathing without any warning. This had been very sudden and unexpected. They had given him a small amount of milk for the first time just a few minutes after, but they didn't think that was relevant. There were several people present, with at least two or three doctors and several nurses.

"I contacted [Father of Baby C], who was at home in bed, and I told him what had happened, and that he needed to come to the hospital urgently.

"Initially I didn't really take in what was happening and I didn't take in the severity, until I was asked by a neonatal nurse, somewhat unexpectedly, whether I wanted someone to call a priest. I remember feeling quite shocked and I asked her if she thought he was going to die, to which she responded, 'Yes, I think so'. This was a nurse I had never met before and I was surprised that this piece of information and communication came from a nurse rather than a doctor.

I am unsure whether she did this on the direction of a doctor or not.

"The nurse did not tell me her name. She was in her mid to late 20s with a fair complexion and brown hair tied back in a ponytail.

"I had to make a decision about calling the priest in the absence of my husband, who had not yet arrived.

Both Roman Catholic and Church of England priests were contacted in order for [Baby C] to be baptised. I had assumed [Baby C] could be baptised by both denominations and had been reassured by the nurse that it was fine to have both. It transpired, once

he had been baptised, he could not then be baptised again into another denomination.

"Whilst waiting for the priest, [Father of Baby C] arrived on the unit. It seemed to take forever for the priest to arrive and all the time we were waiting CPR was being carried out on [Baby C]. We had to wait for around 50 minutes. If I had known it was going to take that long, I wouldn't have prolonged things. I just felt this was my only opportunity to have [Baby C] baptised.

I didn't want to regret not having done it at a later date. It was a way in which we were able to validate that he had been here. We wouldn't have anything else.

[Baby C] was baptised by a Church of England priest.

"After the baptism, [Baby C] was handed to me and I was told he had passed away. I was holding him and he was still breathing. It was awful. I was telling them, 'He's breathing, he's got a pulse, he's not dead'.

John Gibbs said to me, 'What would you like us to do?

Do you want us to carry on?' I said, 'No, I just want us to be together.' He wasn't the same baby now.

"In amongst all of this, they had scanned his brain and it was evident there had been extensive changes as he had been deprived of oxygen for quite a long period of time.

"I'm not sure what the time was. I went to [Baby C] at about 23.20 hours. The CPR and baptism went on for about an hour, so it was probably about half past midnight when we called our parents. Both mine and [Father of Baby C]'s parents came to the hospital, and we went into the lounge at the back of the neonatal unit where we just held [Baby C]. [Baby C] was given some morphine for comfort and he was much more settled with this:

"There were two neonatal nurses that were with us throughout this time. One of them was the nurse who had asked me earlier whether I wanted them to call the priest. The other I have no prior recollection of but I think her first name was Mel. These two nurses attended to us for the remainder of our time there.

They came back to check on our welfare and they also took [Baby C]'s foot and handprints for us to keep. They asked if we wanted a lock of his hair and one of the two nurses took this for us. I cannot remember which nurse this was.

"Shortly before he passed away, [Baby C] was given another dose of morphine by one of the nurses. Again, [Father of Baby C] can't recall which. This was around 05.30 hours on Sunday the 14th."

Moving on to the next day:

"On Thursday, 15 June, I was contacted by either Alder Hey Children's Hospital or the coroner's office.

I was told [Baby C] had been transferred to Alder Hey and they were going to conduct a post-mortem there. I was contacted again later that day to say the post-mortem had now concluded and I should register [Baby C]'s death as natural causes.

"I was told they were still awaiting the results of further tests for a more thorough conclusion, but that could take a few months. [Father of Baby C] and I were then able to arrange [Baby C]'s funeral. We had [Baby C] cremated at Blacon Crematorium."

That concludes the relevant part of [Mother of Baby C]'s statement.

Witness statement of [Father of Baby C] (read)

MR DRIVER: Moving to a statement of her husband, [Father of Baby C], dated 10 December 2019, he identifies himself and states that he married [Mother of Baby C] on 31 May 2013, and that his wife became pregnant in early December 2014.

He states that:

"[He] would describe [Mother of Baby C]'s pregnancy as highly stressful. [He] is not aware of the names of any of the medication or medical procedures she had had to take, leading up to our baby's birth, but I do remember that she had to check her blood pressure on a regular basis.

"Some time before our baby was born [he is not certain on the exact day, but 1 to 2 weeks before [Baby C] was born], [Mother of Baby C]'s blood pressure had increased. As this was one of the symptoms she had been asked to look out for, she attended the Countess hospital so she could be regularly monitored. They admitted her as a safety precaution. I cannot remember exact details of how I was informed. I think [Mother of Baby C] was admitted to hospital following a scan. She stayed at the hospital and was monitored.

"The elective section was planned for the morning of 10 June 2015. On my arrival to the hospital I met up with [Mother of

Baby C] and we were briefed by her consultant, Mr Jim McCormack, and approximately four to five other people.

"We were then led into the operating room and our son, [Baby C], was born by caesarean section shortly afterwards. After being born he was wrapped up and placed into an incubator. I remember the medical staff being really pleased with him as he was displaying positive signs that he was going to be okay. [Baby C] was then taken to the neonatal ward while [Mother of Baby C] was taken to a separate ward to recover.

"I followed [Baby C] to the neonatal ward, which I would describe as a secure ward within the Countess hospital. To gain access you have to be buzzed in by an intercom system. Once inside, there's a corridor with rooms off. I was directed to a glass-fronted large room on the left. This room was able to accommodate other babies but I can't remember if there were other babies present at this time.

"I remained with [Baby C], and [Mother of Baby C] joined us a few hours later. He was still doing well. For the majority of time he needed to use a respirator whilst he was in his incubator, but this wasn't all of the time. [Baby C] would fight with the tubes and face mask. He was capable of breathing independently. Staff continued to monitor his progress. At no time did they express any concern of him not surviving.

"[Baby C] was cared for by a number of nurses whilst at the Countess of Chester Hospital. As I was generally only there during the day, leaving around 7 to 8 pm each evening, I can only recall one specific nurse. I cannot recall the name, which cared for him during the day shifts. I would describe her as Nurse 1."

And he gives this general description:

"White with blonde/brown hair which was tied up in a ponytail, aged mid 20s. She had a mild manner, 5 foot 6 inches tall, slender build. I cannot her name.

I think she was wearing scrubs most of the time, but I cannot be certain.

"As with any premature baby, [Baby C] was not fed initially with any milk as they were monitoring his bowel function, which appeared to be working okay as he was able to pass urine and small poos. On the 11th and 12 June, [Mother of Baby C] and I took turns in holding him.

"On 13 June 2015 I spent the day with [Mother of Baby C] and [Baby C] on the ward. I think this was the day [Baby C] had his

first milk feed. As far as I am aware, no problems arose from this feed and he continued to progress well and staff remained positive. He was still in an incubator and receiving care by staff. Some time early evening, around 7 pm, I left the ward and went home.

"Later that evening I was awoken by my mobile ringing. It was [Mother of Baby C]. She told me that I had to get to the hospital right away, so I grabbed some clothes and immediately made my way.

"As I approached the neonatal ward, [Mother of Baby C] was waiting for me in the corridor. She told me that [Baby C] had collapsed and had stopped breathing. She'd also agreed for a priest to come so that [Baby C] could be baptised as they didn't believe he was going to make it.

We walked into the ward together and entered [Baby C]'s room, which was the same one he'd always been in.

"[Baby C] was lying in his incubator. A plastic casing which that normally encased it had been removed and a red-haired female -- I'm not sure whether she was a nurse a doctor or a midwife. This female, who I had not seen before, was attempting to resuscitate him by performing CPR. Also present was a male consultant.

"The red-haired female, who I will refer to for the purposes of this statement as Nurse 2, attempted to resuscitate [Baby C] for a long time. At one point she turned to the consultant present and said she didn't think she could do anymore. However, she kept going whilst we waited for the priest. Once [Baby C] was baptised, they stopped trying to resuscitate him. It was during that time that [Baby C] started breathing on his own.

"[Mother of Baby C] and [I] were escorted into a family room by two further nurses who I will refer to as Nurses 3 and 4.

There were more people around but I can't picture them.

This included the priest.

"Nurse 3 I describe as a white female with long mousy-brown hair which had been tied up, slim to medium build, aged around mid 20s. I have seen this nurse's picture in the newspaper after her arrest and I now know her name to be Lucy Letby.

"Nurse 4 was also a white female with light brown shoulder length hair, slender build, aged late 20s to early 30s.

"Whilst in this room, [Mother of Baby C] and I took turns holding him as we knew he wasn't going to make it. We just wanted to spend this time together. We'd also called our parents who attended. The priest left shortly after [Mother of Baby C]'s parents' arrival.

"I remember during this time one of the nurses administered some morphine. I think this happened on more than one occasion, possibly two to three times.

"At regular intervals Nurses 3 and 4 came in to check on our welfare. They also took [Baby C]'s hand and footprints for us to keep. Clothes and toys from his cot were also gathered and given to us. There was nothing else that could be done for [Baby C]. It was just a matter of time before he would die, so we just wanted to cuddle him and make sure he was pain-free until that time came.

"I remember at one point one of the nurses -- I think it could have been Nurse 3, but I'm not 100% sure -- came in with a ventilated basket. She turned to us and said words similar to, 'You have said your goodbyes now, do you want me to put him in here?' referring to the basket. This comment shook us. [Mother of Baby C] said, 'He's not died yet'. The nurse then attempted to backtrack and defuse the situation, but I couldn't believe what she'd said. We didn't want to leave him when he was still alive.

"When [Baby C] sadly died it was around 5 am on 14 June. [Baby C] was put in a ventilated basket before being moved to the room next door where we were escorted by either Nurse 3 or 4. Once inside we placed the basket on the floor and said our goodbyes before shutting the door and leaving him. [Mother of Baby C] then went back to the maternity ward to collect her belongings and to say goodbye to the staff there."

That concludes, my Lord, the relevant parts of [Father of Baby C]'s statement.

MR JUSTICE GOSS: Yes, thank you very much.

MR JOHNSON: Can I call Kate Tyndall, please.

MR JUSTICE GOSS: Yes.

MRS KATE TYNDALL (recalled)

MR JOHNSON: My Lord, we're going to adopt the same approach in every single case. So we will start with the sequence of events.

Examination-in-chief by MR JOHNSON

MR JOHNSON: Mrs Tyndall, are you responsible for compiling these sequence of events charts in the case of [Baby C].

TYNDALL: Yes, I am.

JOHNSON: I would like to take you through this document, please.

Just so the jury understands, my Lord, we're going to deal this a little bit more quickly than we did in the case of the [Surname of Babies A & B] children because the jury will now understand how the system works. So we're not going to go behind many of these tiles where there are witnesses to come who will deal with them in detail. So we won't have that repetition that we've had in previous cases.

MR JUSTICE GOSS: All right. You understand what Mr Johnson is saying? Because that was to introduce you to the scheme of things. Now you're familiar with how the sequence of events works and you know the documents behind the summaries, and that's what the witnesses will speak to when they come to give their evidence. It's just to take you through the sequence. Literally the purpose of it is to sequence the events.

MR JOHNSON: Thank you, my Lord.

So from time to time, as we saw with the sequences for the [Surname of Babies A & B] twins, events relating to other children appear to contextualise what was going on; is that right?

TYNDALL: Yes, that's correct.

JOHNSON: And so, for example, we have at tile number 1 in [Baby C]'s sequence [Baby A]'s collapse at 20.26 and his death, which was timed as 20.58?

TYNDALL: Yes.

JOHNSON: And at tile 3, on the next shift, we have the event in which the nasal prongs came out of [Baby B]'s nose?

TYNDALL: Yes.

JOHNSON: Then her collapse at just after midnight at the beginning of the day of 10 June?

TYNDALL: Yes.

JOHNSON: And we see on that screen that, at 15.31 that same day, [Baby C] was born?

TYNDALL: Yes.

JOHNSON: We can see that timed at 15.31 and we can see Dr Chowdhury's notes relating to [Baby C] at 16.00 hours?

TYNDALL: Yes.

JOHNSON: And again we will come to this with the witnesses, but an hour later at 17.00 hours communication between Drs Chowdhury and Ogden with [Parents of Baby C]?

TYNDALL: Yes.

JOHNSON: Some more notes timed at about the same time by Dr Sally Ogden?

TYNDALL: Yes.

JOHNSON: And further notes from Dr Brunton at 6 o'clock and Dr Chowdhury at 6 o'clock?

TYNDALL: Yes.

JOHNSON: Just pausing there for a second, just to remind us all, Dr Brunton a registrar, ST4. Dr Chowdhury, a GP training in his second year of training is that right.

GP ST2?

TYNDALL: Yes.

JOHNSON: So working in the hospital as part of his training.

We then have Christopher Booth's notes written at the end of the day shift, written after the shift had finished at 20.28, but relating to events at 18.00 hours?

TYNDALL: Yes.

JOHNSON: Then a record at tile 12 of a communication Christopher Booth had had, or Nurse Booth had had at 18.30 with the [Surname of Baby C] parents?

TYNDALL: Yes.

JOHNSON: A further update on the computer relating to [Baby C] and a further update relating to the parents --

TYNDALL: Yes.

JOHNSON: -- made little later at 8.50.

Then there is, at tile 15, we have a pink-banded tile. What does that pink band denote?

TYNDALL: So that is to show -- so for the sequence of events, we'd include details of the birth and the baby's admittance to the NNU, and then we would -- in most cases, the sequence of events takes place 24 hours before the collapse of the baby and a period afterwards, and that's to indicate that the baby has been in the neonatal unit from 20.50 hours, which is the timed entry of the tile before, and then there will be a gap in the medical records until 24 hours before the collapse.

JOHNSON: Yes, okay. So this is just a very short summary to cover a period of time leading up to the time before the collapse?

TYNDALL: Yes.

JOHNSON: All right, thank you.

We then have some text messages, I think, and I would like to you take us through these if you would, please, starting at sheet 16.

TYNDALL: Okay. So this is a text message sent from Lucy Letby to Yvonne Griffiths that reads:

"Hi Yvonne, are you okay for staffing over next few days? I don't have anything on if (sic) need any extra or need to change my nights x."

JOHNSON: So that's at 14.08 on the Thursday?

TYNDALL: Yes.

JOHNSON: Does that elicit a response at 17.35?

TYNDALL: Yes. Yvonne Griffiths replies:

"Thank you for your kind offer. We are okay until Saturday night when we only have five staff so we can re-assess then."

JOHNSON: Then within 7 minutes did Lucy Letby respond?

TYNDALL: Yes, she responds with: Okay, think I need to throw myself back in on Sat."

JOHNSON: Thank you. And then Yvonne Griffiths responds within a couple of minutes?

TYNDALL: "Hopefully it may settle down by then."

JOHNSON: And then Lucy Letby?

TYNDALL: "Hope so. But think from a confidence point of view I need to take an ITU baby soon X."

JOHNSON: Yvonne Griffiths' response, please?

TYNDALL: "Yes, it does knock us a bit when things like that happen but it is okay to have time out as well. Enjoy the sun x."

JOHNSON: So just -- we already have seen the context of this.

It's a couple of days after the death of [Baby A] --

TYNDALL: Yes.

JOHNSON: -- and the collapse of [Baby B]? All right.

So then another pink-edged tile at number 22?

TYNDALL: Yes.

JOHNSON: And does that follow the same format as the previous one?

TYNDALL: Yes.

JOHNSON: Thank you. We then come to the nursing handover at the beginning of the shift on -- of the night shift, I should say, on Friday, 12 June 2015 where -- on that shift, I think, Melanie Taylor was the designated nurse; is that right?

TYNDALL: Yes.

JOHNSON: Can we just go behind that time, please, into the document, please. So that's just -- that's just a handover sheet, is it?

TYNDALL: That's the nursing notes.

JOHNSON: Nursing notes, sorry?

TYNDALL: So we take the designated nurse from the nurse that's made the entry that normally starts with "care taken over" or "care handed over".

JOHNSON: Okay. Thank you. So if we can go back, please. Do we then have the -- I'll call them the standard observations. There are charts that we saw in both [Surname of Babies A & B] twins' cases?

TYNDALL: Yes.

JOHNSON: So on the hour here we have readings taken at 8 pm?

TYNDALL: Yes.

JOHNSON: Is that 3 and then at 9 pm?

TYNDALL: Yes.

JOHNSON: And some more clinical notes made by Dr Katherine Davis, registrar, at 21.20?

TYNDALL: Yes.

JOHNSON: Reference to a pathology sample in which the senior house officer, Theresa MacCarrick, was involved?

TYNDALL: Yes.

JOHNSON: And then again the observations on the hour at 10 pm?

TYNDALL: Yes.

JOHNSON: And 11 pm?

TYNDALL: Yes.

JOHNSON: And midnight?

TYNDALL: Yes.

JOHNSON: The nursing notes made by the designated nurse, Melanie Taylor, at 00.03?

TYNDALL: Yes.

JOHNSON: Followed by a record of medication given to [Baby C] at 00.33?

TYNDALL: Yes.

JOHNSON: Are these standard observation charts yet again at 1 o'clock?

TYNDALL: Yes.

JOHNSON: Blood gases at 01.50?

TYNDALL: Yes.

JOHNSON: Observations again at 2 o'clock?

TYNDALL: Yes.

JOHNSON: And 3 o'clock?

TYNDALL: Yes.

JOHNSON: An update from the designated nurse, Melanie Taylor, at -- relating to events at 03.09?

TYNDALL: Yes.

JOHNSON: And then the -- the early observations at 4 am?

TYNDALL: Yes.

JOHNSON: And 5 am?

TYNDALL: Yes.

JOHNSON: An update on the nursing notes, again Melanie Taylor, at 5.27?

TYNDALL: Yes.

JOHNSON: Observations again at 6?

TYNDALL: Yes.

JOHNSON: Medication at 6?

TYNDALL: Yes.

JOHNSON: And 6.39?

TYNDALL: Yes.

JOHNSON: Observations at 7?

TYNDALL: Yes.

JOHNSON: Then we get to the handover to the day shift on Saturday, 13 June, at 7.30 in the morning?

TYNDALL: Yes.

JOHNSON: So we're now within 24 hours of [Baby C]'s death which occurred on the night shift, Saturday into Sunday; is that right?

TYNDALL: Yes.

JOHNSON: Thank you. So Joanne Williams, the designated nurse, taking hourly readings as we have seen in the cases of the [Surname of Babies A & B] twins?

TYNDALL: Yes.

JOHNSON: Blood gas just before 9 am followed by standard observations?

TYNDALL: Yes.

JOHNSON: At 9.30 am, the consultant, Dr Gibbs, doing a ward round with the registrar, Dr Sally Ogden?

TYNDALL: Yes.

JOHNSON: And making observations and recording those observations in the medical notes?

TYNDALL: Yes.

JOHNSON: Thank you. 10 o'clock, the standard observations?

TYNDALL: Yes.

JOHNSON: And 11 o'clock?

TYNDALL: Yes.

JOHNSON: We then have some nursing notes which were written later in the shift which relate to this time but which we'll deal with in the context of the evidence of Joanne Williams.

TYNDALL: Okay.

JOHNSON: The standard observations taken at midday?

TYNDALL: Yes.

JOHNSON: Medication at midday?

TYNDALL: Yes.

JOHNSON: Sodium chloride given at 12.30 hours?

TYNDALL: Yes.

JOHNSON: And the standard observations again at 13.00 hours?

TYNDALL: Yes.

JOHNSON: There is a prescription chart or two prescription charts, or infusion prescription charts, I should say, which are signed by Yvonne Griffiths and [Nurse B] --

TYNDALL: Yes.

JOHNSON: -- at 1 pm?

TYNDALL: Yes.

JOHNSON: And Joanne Williams making a reference in the notes to this time in notes that she made later in the day?

TYNDALL: Yes.

JOHNSON: Ranitidine given at 1 pm by Ailsa Simpson and Joanne Williams?

TYNDALL: Yes.

JOHNSON: And at 2 pm these standard observations taken again?

TYNDALL: Yes.

JOHNSON: Blood gases at 14.45?

TYNDALL: Yes.

JOHNSON: Followed by nursing notes which were written up later?

TYNDALL: Yes.

JOHNSON: And 5 minutes later, a pathology sample collected by Joanne Williams, the nurse?

TYNDALL: Yes.

JOHNSON: Ten minutes later, nursing notes relating to an event which was noted up an hour and 19 minutes later?

TYNDALL: Yes.

JOHNSON: And then again followed by the hourly observation charts at 3 pm?

TYNDALL: Yes.

JOHNSON: Thank you.

We then come to an ultrasound of [Baby C] taken at 15.45 that afternoon?

TYNDALL: Yes.

JOHNSON: We haven't seen one of these before. It has the yellow edging which relates to scans and X-rays, doesn't it?

TYNDALL: Yes.

JOHNSON: Thank you. Given that we haven't seen one, if we could just go behind this one, please, to the -- no. So -- well, for those that understand these things, no doubt very informative, but for the rest of us, perhaps not so informative. All right, thank you.

So we then have notes made in the medical records at 15.55 by Dr Gibbs, the consultant?

TYNDALL: Yes, about the ultrasound.

JOHNSON: About the ultrasound, thank you. And then the observation chart for 4 pm?

TYNDALL: Yes.

JOHNSON: Thank you. And further nursing notes -- this is when the nursing notes were made relating to events earlier in the day, as we have seen them marked out as we've gone along?

TYNDALL: Yes, this is when she seems to write up her nursing notes for everything that's happened on the shift.

JOHNSON: Thank you. And then the standard observations taken again at 5 pm?

TYNDALL: Yes.

JOHNSON: Further on in the shift now, at 17.32, Joanne Williams making a record of communications that she had had with [Parents of Baby C]?

TYNDALL: Yes.

JOHNSON: And then the standard observations taken by a different nurse, this time Ailsa Simpson --

TYNDALL: Yes.

JOHNSON: -- at 6 pm?

TYNDALL: Yes.

JOHNSON: At just after 19.40 that same evening, so as we're getting towards the end of the -- well, this is during the handover, isn't it, between 7.30 and 8 pm, Joanne Williams making up her notes for the balance of the shift?

TYNDALL: Yes. Yes. So that's anything that happened between her making her notes at 16.19 and the --

JOHNSON: Sorry, yes?

TYNDALL: -- 18.00.

JOHNSON: Those actually relate to events at 6 pm?

TYNDALL: Yes.

JOHNSON: Sorry, I have slightly misled you there.

At 6.15 pm, more blood gases?

TYNDALL: Yes.

JOHNSON: And more medication for [Baby C] at 18.26 and 18.27?

TYNDALL: Yes.

JOHNSON: At 18.29, some caffeine?

TYNDALL: Yes.

JOHNSON: And then, at 18.45, gentamicin?

TYNDALL: Yes.

JOHNSON: Followed by the standard observations at 7 pm?

TYNDALL: Yes.

JOHNSON: We then see swipe data relating to the arrival of the nursing staff for the night shift on Saturday, 13 June?

TYNDALL: Yes, that's correct.

JOHNSON: So first in was Elizabeth Marshall, followed by Melanie Taylor?

TYNDALL: Yes.

JOHNSON: Sophie Ellis?

TYNDALL: Yes.

JOHNSON: [Nurse B]?

TYNDALL: Yes.

JOHNSON: Nicola Dennison?

TYNDALL: Yes.

JOHNSON: There is a nursing record. Is this the final update for Joanne Williams for that shift?

TYNDALL: Yes, I think so. Yes.

JOHNSON: Yes. And then the shift handover itself is noted in the timeline at tile 138. Is there a document behind that?

TYNDALL: There should be the staffing chart.

JOHNSON: We will look at that, please. Could we go behind 138?

So this is what the jury have seen before, a record of some of the information they've already seen in this sequence, namely that Dr Gibbs was working that Saturday, and the night registrar is identified as Katherine Davis, the night SHO. The shift leader being [Nurse B], Sophie Ellis, Melanie Taylor, both working with Lucy Letby, and the nursery nurse is Nicola Dennison and Elizabeth Marshall.

Do we then have a list of the children that were in the unit for this night shift?

TYNDALL: Yes.

JOHNSON: We can see, at number 1, [Baby C], and his designated nurse for the shift was Sophie Ellis?

TYNDALL: Yes.

JOHNSON: We can see the name of [Baby B] there at number 4, and we see the other 10 children identified by initials, and notably Lucy Letby was the designated nurse for JE?

TYNDALL: Yes.

JOHNSON: Who is number 3, and PE, who is number 10?

TYNDALL: Yes.

JOHNSON: Thank you. Do we then have nursing notes written by Sophie Ellis which relate in part at least to the handover?

TYNDALL: Yes. So these are the notes that talk about how [Baby C] was at the start of the shift when she took over his care.

JOHNSON: Yes. We can see from the tile itself that the notes were written at about 8.45 on the Sunday morning?

TYNDALL: Yes.

JOHNSON: Which of course would have been after the shift had ended. And the shift on which [Baby C] had died?

TYNDALL: Yes.

JOHNSON: Thank you. So moving on, please, do we have the general readings as we have seen in so many other -- on so many other occasions taken at 8 pm?

TYNDALL: Yes.

JOHNSON: At 9 pm?

TYNDALL: Yes.

JOHNSON: And this of course, as we can see, is a WhatsApp text; is that right?

TYNDALL: Yes.

JOHNSON: Just -- we've seen that Lucy Letby was on duty that night?

TYNDALL: Yes.

JOHNSON: That information is taken from the staff rota; is that right?

TYNDALL: Yes.

JOHNSON: And also from the medical records on the unit that night?

TYNDALL: Yes.

JOHNSON: So although there's no swipe data for her coming in, she was there?

TYNDALL: Yes.

JOHNSON: So could you take us through the text messages, please, from 146?

TYNDALL: So at 21.29 Jennifer Jones-Key sends a message to Lucy Letby which reads:

"You okay? x."

JOHNSON: Thank you.

TYNDALL: Lucy Letby replies at 21.40 saying:

"I just keep thinking about Mon. Feel like I need to be in 1 to overcome it but [Nurse B] said no."

JOHNSON: So Monday that had been -- what, the 8th -- the 13th?

Yes, Monday would have been 8 June?

TYNDALL: Yes, Monday was 8 June.

JOHNSON: And that was the day that [Baby A] had died?

TYNDALL: Yes.

JOHNSON: [Baby A] had been in room number 1 when he'd died?

TYNDALL: I believe so.

JOHNSON: Yes. Thank you. On to the next one, please.

TYNDALL: Jennifer Jones-Key replies at 21.51:

"I agree with her. Don't think it will help. You need a break from full on ITU. You have to let it go or it will eat you up. I know not easy and it will take time x."

JOHNSON: Thank you.

TYNDALL: Lucy Letby replies at 21.55:

"Not the vented baby necessarily. I just feel I need to be in 1 to get the image out of my head. Mel said the same and [Nurse B] let her go. Being in 3 is eating me up. All I can see is him in 1."

JOHNSON: Thank you.

TYNDALL: Lucy Letby then follows this up with another message at 21.56 that reads:

"It probably sounds odd but it's how I feel."

Jennifer Jones-Key replies at 21.57:

"Well it's up to you but don't think it's going to help. It sounds very odd and I would be complete opposite. Can understand [Nurse B]. She is trying to look after you all."

JOHNSON: Thank you.

TYNDALL: Then Lucy Letby replies at 21.51:

"Well that's how I feel, from when I've experienced it at Women's I have needed to go straight back and have a sick baby otherwise the image of the one you lost never goes. Why send Mel in if she's trying to look after us? She was in bits over it."

And then follows this up at 22.00 with a message:

"Don't expect people to understand but I know how I feel and how I've dealt with it before. I've voiced that so can't do anymore but people should respect that."

JOHNSON: Thank you.

We then have the observations of [Baby C] taken by Sophie Ellis at 10 pm?

TYNDALL: Yes.

JOHNSON: Followed by Jennifer Jones-Key's response to Lucy Letby at 22.08?

TYNDALL: Yes.

JOHNSON: That just reads "Okay"?

TYNDALL: Yes.

JOHNSON: Do we then have some swipe data relating to Dr Davis?

TYNDALL: Yes.

JOHNSON: Thank you. And a continuation of the WhatsApp conversation between Jennifer Jones-Key and Lucy Letby?

TYNDALL: Yes. Jennifer Jones-Key replies at 22.29:

"I think they do respect it but also trying to help you. Why don't you go in 1 for a bit?"

Lucy Letby replies at 22.30:

"Yeah, I have done couple of meds in 1. I'll be fine."

Jennifer Jones-Key replied at 22.35:

"It didn't sound like you would be? Sorry, was eating my tea."

Then Lucy Letby replies at 22.38:

"Forget I said anything, it will be fine. It's part of the job. Just don't feel like there is much team spirit tonight."

JOHNSON: Okay. Now, above that message, in italicised font, is:

"Private content relating to third party not relevant."

Has the content of that message been served on the defence as part of the evidence in the case?

TYNDALL: Yes.

JOHNSON: Thank you. So can we move on, please.

TYNDALL: Then Jennifer Jones-Key replies at 22.41:

"I'm not going to forget but just think your way too hard on yourself. It is part of the job but the worst part, but I do believe it makes us stronger people."

JOHNSON: Thank you. We then see some more swipe data of the registrar, Dr Davis?

TYNDALL: Yes.

JOHNSON: And followed by a continuation of the WhatsApp conversation between Lucy Letby and Jennifer Jones-Key?

TYNDALL: Yes. Lucy Letby replies at 22.53:

"Unfortunately I've seen my fair share at the Women's, but you are supported differently and here it's like people want to tell you how to think/feel. Anyway.

Onwards and upwards. Just shame I'm on with Mel and [Nurse B]. Sophie in 1 so haven't got her to talk to either."

JOHNSON: Thank you.

TYNDALL: And Jennifer Jones-Key replies at 22.57:

"Work is work. A lot of the girls say Women's don't support and tell them to get on with it. I think they don't mean to tell you though and were over-caring sometimes. Yeah, that's not good but you've got Liz."

JOHNSON: Thank you.

Do we then have a note relating to 11 pm?

TYNDALL: Yes.

JOHNSON: That's written by Sophie Ellis at the end of the shift?

TYNDALL: Yes.

JOHNSON: And an addendum note as well that we have some more information relating to 11 pm?

TYNDALL: Yes.

JOHNSON: Thank you. And do we then have the standard observations taken at the same time --

TYNDALL: Yes.

JOHNSON: -- on the hour, followed by a continuation of the WhatsApp conversation?

TYNDALL: Yes. So at 23.01 Lucy Letby replies to Jennifer Jones-Key:

"Women's can be awful but by I learnt hard way that you have to speak up to get support. I lost a baby one day and few hours later was given another dying baby just born in the same cot space. Girls there said it was important to overcome the image. It was awful but by end of day I realised they were right. It's just different here."

Then follows this up with a message at 23.02:

"Anyway, forget it. I can only talk about it properly with those who knew him and Mel not interested so I'll overcome it myself. You get some sleep."

Jennifer Jones-Key replies at 23.05:

"That's a bit mean, isn't it? Don't have to know him to understand. We have all been there. Yep, off to bed now."

JOHNSON: Thank you. We then have swipe data relating to Nicola Dennison at 23.07?

TYNDALL: Yes.

JOHNSON: Followed by a continuation of the text conversation between Lucy Letby and Jennifer Jones-Key?

TYNDALL: Yes, Lucy Letby replies, 23.09:

"I don't mean it like that, just that only those who saw him know what image I have in my head."

And then follows it up with:

"Forget it. I'm obviously making more of it than I should. Sleep well xx."

JOHNSON: Thank you. More swipe data relating to Dr Hor?

TYNDALL: Yes.

JOHNSON: Followed by nursing records relating to this time but which were made at -- a few days later?

TYNDALL: A few days later.

JOHNSON: On 16 June?

TYNDALL: Yes.

JOHNSON: Let's just deal with this because this is out of the pattern that we've seen before. So these are notes made a few days later?

TYNDALL: Yes.

JOHNSON: Can we just see what they relate to, please. Could you just give us your summary, please?

TYNDALL: Yes. So:

"Had 2x fleeting bradys, self-correcting, not needing any intervention, shortly before prolonged brady and apnoea requiring resus. Minimal bile aspirates gained. Green in colour. NGT aspirated before first feed. Bile aspirated."

JOHNSON: Thank you. Then back, please. Do we then have a record of the events concerning [Baby C]?

TYNDALL: Yes.

JOHNSON: We then have nursing notes written up the following day by both Sophie Ellis and Melanie Taylor?

TYNDALL: Yes.

JOHNSON: Swipe data followed by clinical notes written by Dr Katherine Davis, the registrar?

TYNDALL: Yes.

JOHNSON: And Dr Hor written -- who is the house officer, written at 1.30 am, but relating to events at 23.26?

TYNDALL: Yes, it's Dr Davis that makes those notes. Dr Hor is in there because he's noted in her notes as attending with her.

JOHNSON: Thank you --

MR JUSTICE GOSS: You're getting softer and softer.

TYNDALL: Sorry.

MR JOHNSON: It's getting very warm in here.

MR JUSTICE GOSS: We are not going to go on much longer.

MR JOHNSON: I think I'm dropping my voice as well, Dr Davis, her swipe data relates to 23.28?

TYNDALL: Yes.

JOHNSON: Retrospective nursing notes written towards the end of the shift, but relating to these events, made by Melanie Taylor?

TYNDALL: Yes.

JOHNSON: Clinical notes written concerning [Baby C]'s collapse by the on-call consultant, Dr Gibbs?

TYNDALL: Yes.

JOHNSON: And we see what the jury will recognise now as the -- really the drug of last resort in a resuscitation, adrenaline, timed at 23.30 and 23.33?

TYNDALL: Yes.

JOHNSON: Together with a bleep record at the same time?

TYNDALL: Yes.

JOHNSON: Followed by another drug of last resort, as referred to by Dr Jayaram the other day, sodium bicarbonate?

TYNDALL: Yes.

JOHNSON: Followed by yet another bleep record at 23.35?

TYNDALL: Yes.

JOHNSON: Followed by more notes made by Dr Gibbs?

TYNDALL: Yes.

JOHNSON: And swipe data. This presumably is him coming into the unit, is it, at 23.36?

TYNDALL: Yes.

JOHNSON: We see three entries as a matter of fact. Is there an obvious explanation for why there were three separate --

TYNDALL: It's just showing how he -- one of them, I believe, is him coming into the hospital, one into the building, I think, and one into the -- the last one is him into the neonatal unit itself.

JOHNSON: So it's the progress from outside to the neonatal unit itself?

TYNDALL: Yes.

JOHNSON: So three doors. Thank you.

Again, adrenaline at 23.36 with the names of the nurses who checked the dose.

Dextrose at 23.38, more bicarbonate of soda at 23.38?

TYNDALL: Yes.

JOHNSON: Followed by the registrar, Dr Katherine Davis' notes, written retrospectively at 1.30?

TYNDALL: Yes.

JOHNSON: And yet another bolus of adrenaline at 23.39?

TYNDALL: Yes.

JOHNSON: Saline at the same time followed by another injection of adrenaline at 23.41?

TYNDALL: Yes.

JOHNSON: And at 23.43?

TYNDALL: Yes.

JOHNSON: More saline at 23.47?

TYNDALL: Yes.

JOHNSON: Yet more adrenaline at 23.52?

TYNDALL: Yes.

JOHNSON: Calcium gluconate at 23.52?

TYNDALL: Yes.

JOHNSON: Followed by more notes made by the consultant, Dr Gibbs?

TYNDALL: Yes.

JOHNSON: And notes of Melanie Taylor?

TYNDALL: Yes.

JOHNSON: Yet more notes of Dr Gibbs at 00.45?

TYNDALL: Yes.

JOHNSON: We then move on an hour -- about half an hour or so, 37 minutes, to Dr Davis' swipe data?

TYNDALL: Yes.

JOHNSON: Eight minutes later, her clinical notes are updated?

TYNDALL: Yes.

JOHNSON: And more notes at 2 am from Dr Davis?

TYNDALL: Yes.

JOHNSON: Then we have more nursing notes made by Mel Taylor?

TYNDALL: Yes.

JOHNSON: And a morphine injection or -- I should say a dose of morphine given to [Baby C], signed for by Nurses Taylor and [Nurse B]?

TYNDALL: Yes.

JOHNSON: More swipe data from Dr Davis at 2.28 followed by some clinical notes she made at 5 am?

TYNDALL: Yes.

JOHNSON: Retrospective -- a retrospective nursing notes made by Melanie Taylor, times two?

TYNDALL: Why.

JOHNSON: Which include some that she made actually in the early hours of the following morning, the 15th. So that would be the end of the following night shift, would it?

TYNDALL: Yes, but they're about the events on -- at that time.

JOHNSON: Yes. Again, we will come to all this in the evidence, but some more, morphine for [Baby C] at 5.10?

TYNDALL: Yes.

JOHNSON: Swipe data relating to Melanie Taylor at 05.21 and swipe data relating to Dr Davis at 05.53?

TYNDALL: Yes.

JOHNSON: And then something we've not seen before:

"Incident reported on DATIX."

So we will just deal with that. If we could go behind that time, please. Could we go to the document itself, please.

Can we scroll down, please. All right.

Now, is this a type of incident report -- it's not your document, it's something produced by the hospital, but essentially does it give the identity of the person to whom the incident relates?

TYNDALL: Yes.

JOHNSON: And then does it set out, in a pro forma setting, an explanation of the incident itself?

TYNDALL: Yes.

JOHNSON: All right. Well, we won't deal with the detail of it now. But we can see that that is the second of nine pages. So the document itself is eight pages long?

TYNDALL: Yes.

JOHNSON: Okay. Thank you.

[Baby C]'s death was formally certified as having occurred at 5.58 on Sunday, 14 June?

TYNDALL: Yes.

MR JUSTICE GOSS: Mr Johnson, I'm looking to see how many more slides there are to go through, and we're running out of time.

MR JOHNSON: Yes.

MR JUSTICE GOSS: I'm just wondering whether that, although it's a little bit early, I think that might be a good point to stop.

MR JOHNSON: Yes.

MR JUSTICE GOSS: All right. And then we will continue with the sequence of events tomorrow morning. All right?

Thank you very much. You can come back tomorrow?

TYNDALL: Yes.

MR JUSTICE GOSS: Thank you very much indeed.

Members of the jury, 10.30 tomorrow morning, please, and of course remember your obligations. No communication by any means with anyone and no research in relation to anything to do with this case or anyone involved in it.

Thank you very much indeed.

(In the absence of the jury)

MR JUSTICE GOSS: Sorry, could you just pause a moment.

MR ASTBURY: Thank you, my Lord. Apologies, but we've received rather a last minute special measures application again for the second time. I do apologise.

This time the statement is ready and it's been provided to the defence. Just to flag it up. I'm happy to make it in the morning.

MR JUSTICE GOSS: If you give me a copy and I'll --

MR ASTBURY: It's a witness we will get to at some point tomorrow, we expect. It's for screens. I don't know if my learned friend Mr Myers KC has room to digest it.

MR JUSTICE GOSS: All right. We will deal with it in the morning then.

MR MYERS: My Lord, there's no requirement to delay the decision if your wishes. The position with these witnesses generally, unless you say otherwise, is that, as we set out before, we recognise it's in the purlieu of the court to grant the measure.

We do note -- I don't want to read through all of this now, but the final lines on the first page, the reason for it appears to be an emotional reaction, which is phrased in a slightly different way from the way we normally see applications for special measures, although it's connected with the assertion it will have a detrimental effect on her ability to communicate evidence to the court. Well, that's the test.

MR JUSTICE GOSS: It's indicating she won't be able to give her best evidence.

MR MYERS: No, it indicates that. So whilst we observe with some consternation the reason given for it with what follows, it's in the court's hands. It's not something we can object to, and our position is as it has been with these applications to this point.

MR JUSTICE GOSS: Yes. Right. Well, I'm satisfied that the statutory criteria are met for the granting of the special measures direction that is sought, namely that this witness may be screened from others in court.

I will actually -- when the next witness who is screened comes in, I'll just remind the jury that this is a constant pattern. I'm sure -- say to them I'm sure that they are following my instruction to attach no significance whatsoever to the screen.

MR MYERS: Yes, we are anxious for that because the concern of the defence is always that there might be something read into it which doesn't follow at all.

MR JUSTICE GOSS: No. Well, experience, certainly my experience has been, because I have granted many of these measures, is that by the time I get to the end of the case, I have said it so many times and I say it to them in the summing up as well in writing in a direction, and they just look at me as though we'd effectively forgotten about screens in any event. But I will do it because it should be done.

MR MYERS: We are grateful, my Lord, thank you.

MR JUSTICE GOSS: And it will be done. Thank you very much.

All right. Yes, thank you.

(3.40 pm)

(The hearing adjourned until Thursday, 27 October 2022 at 10.30 am)